

Choosing the Right LLM

How to select, train and apply state-of-the-art LLMs to real-world business use cases



Ed Donner

Co-Founder and CTO, Nebula.io

Starting with an xkcd classic



Today is about **STIRRING THE PILE.**

Thank you for spending
4 hours on Thurs with me

I'll keep it interactive –
please use Q&A & chat



See resources webpage
for links and repo



Setting you up to get the most out of the training

- Bring up the code from the course repo
- After the Live Event, run the code yourself, confirm the results and reach out with questions
- Ask me questions relentlessly! That's what I'm here for.

And...



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And...

I LOVE EMOJIS!!! ❤️👍

Agenda

Segment 1: The LLM Frontier



Agenda

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Segment 2: The Model Match-up



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Segment 1: The LLM Frontier



Segment 2: The Model Match-up



Segment 3: Time To Experiment & Code



Agenda

Segment 1: The LLM Frontier



Segment 2: The Model Match-up



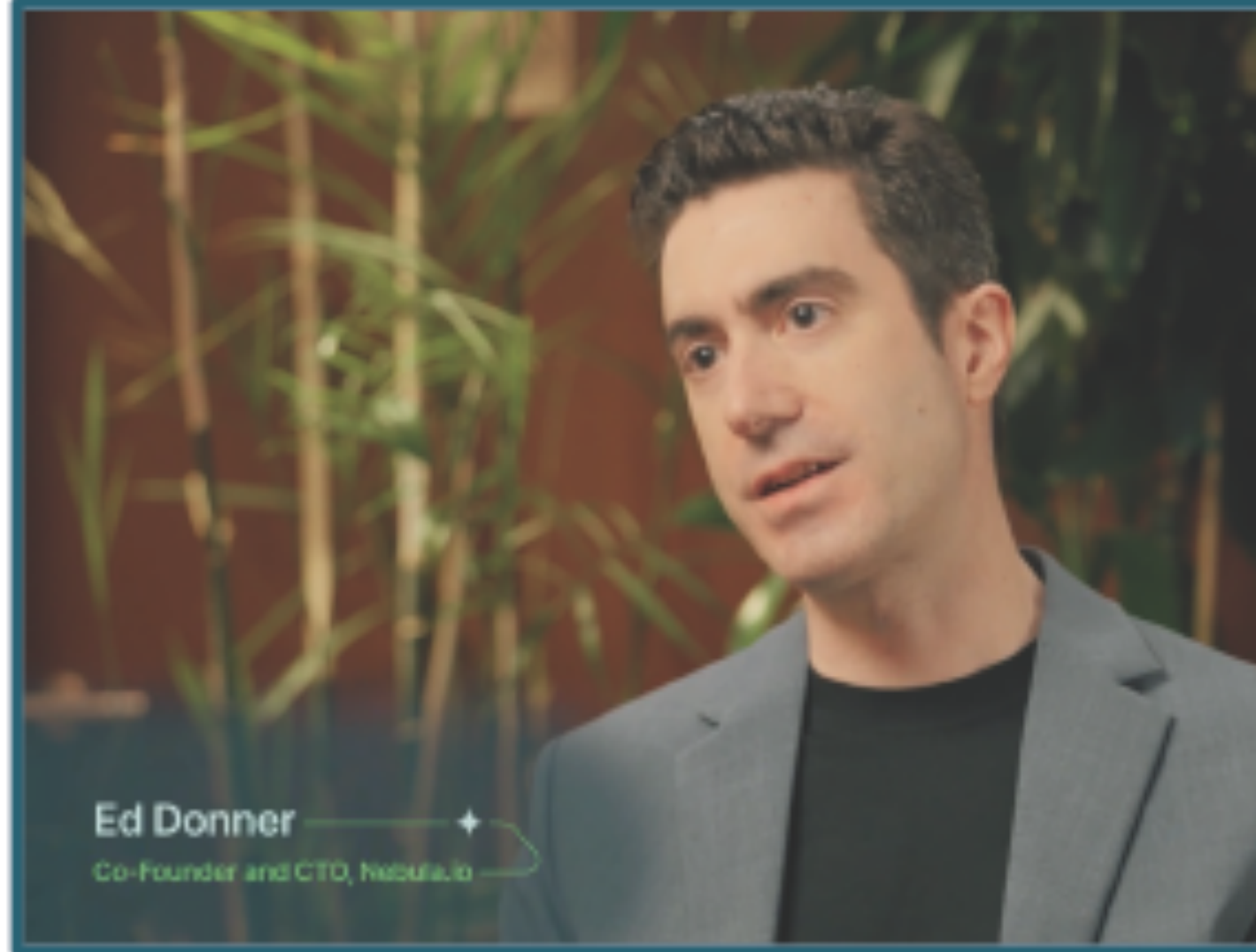
Segment 3: Time To Experiment & Code



*This will equip you with an approach for comparing LLMs and a strategy for applying them to your business problems, so that you can deliver **significant commercial value**.*

Introducing me..

Co-Founder and CTO of Nebula.io; 20-year career as Technology and Data Science leader, entrepreneur and coach



POLL: Introducing YOU

What best describes your experience with LLMs (select one):

A

I've used ChatGPT,
done prompt
engineering

B

I've made API calls
to LLMs

C

I've used techniques
like RAG or LoRA
fine-tuning

D

I've built an Agentic AI
with technologies like
MCP



The Cherry On Top

- Caters for all levels of experience
 - Too hard? Build intuition
 - Too easy? Hang in there...
 - Attended my prior courses? There's more!
- Culminating in great projects
 1. Multi-modal AI Assistant
 2. RAG Question Answerer
 3. QLoRA Fine-tuned LLM
- Featuring Gemini 3.1, GPT 5.4, Claude Opus 4.6, GLM 5, Grok 4.20, Gemma 4
- **NOW – let's get started!**



SEGMENT 1

The LLM Frontier

From Start to State-Of-The-Art

Dizzying Rise of the Generative Pretrained Transformer

Google OpenAI

2017

2018

2019

2020

2021

2022

2023

2024

2025

GPT-1

GPT-2

GPT-3

RLHF and
ChatGPT

GPT4

GPT4o

GPT5

Google scientists
publish seminal
paper "Attention is
All You need"
proposing a new
model architecture
called the
Transformer

Model explosion: an overwhelming number to choose from

- Closed-Source Frontier Models
 - GPT 5.4 from OpenAI
 - Claude 4.6 Opus from Anthropic
 - Gemini 3.1 Pro Preview from Google
 - Grok 4.20 from X
- Open-Source
 - GPT-OSS from OpenAI
 - Gemma 4 from Google
 - GLM 5 from z.ai
 - MiniMax M2.7 from MiniMax AI



Four Powerful Techniques for optimizing LLM performance

Four Powerful Techniques for optimizing LLM performance



Fine-Tuning

Four Powerful Techniques for optimizing LLM performance



Fine-Tuning



Context Engineering

Four Powerful Techniques for optimizing LLM performance



Fine-Tuning



Context Engineering

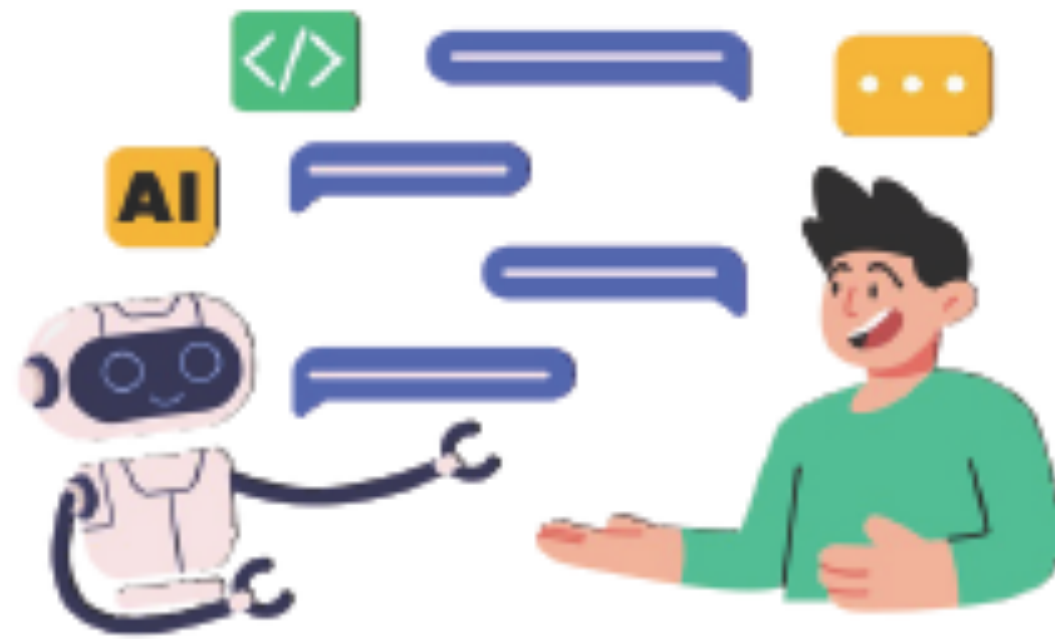


RAG

Four Powerful Techniques for optimizing LLM performance



Fine-Tuning



Context Engineering



RAG



Agentic AI

Four Powerful Techniques for optimizing LLM performance

TRAINING TIME



Fine-Tuning

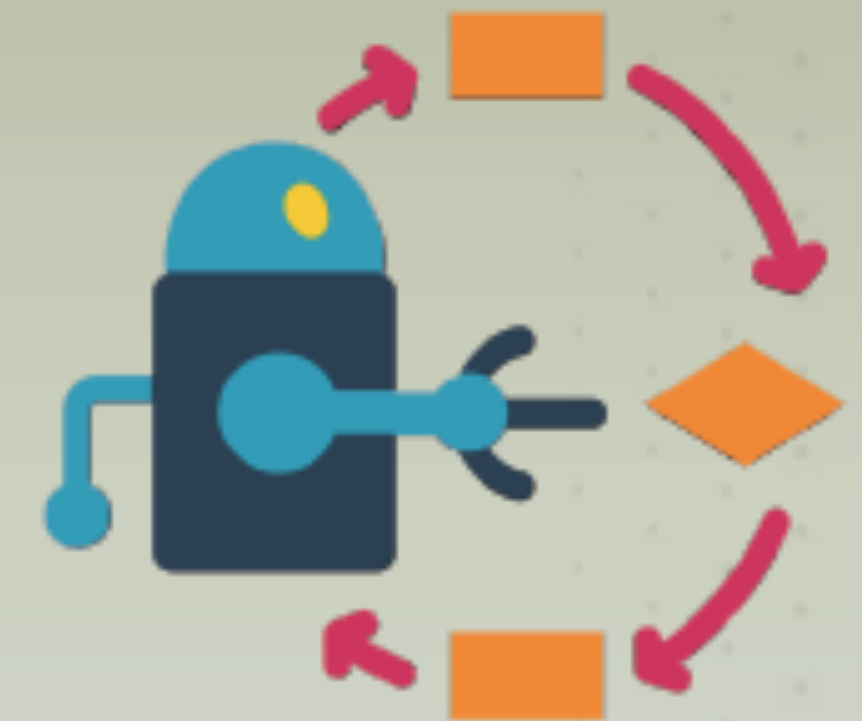
INFERENCE TIME



Context Engineering



RAG



Agentic AI

Tools of the trade

We will be experimenting with these tools and frameworks today.

Please give emojis when I mention ones you like:



Hugging Face



gradio

 **Weights & Biases**

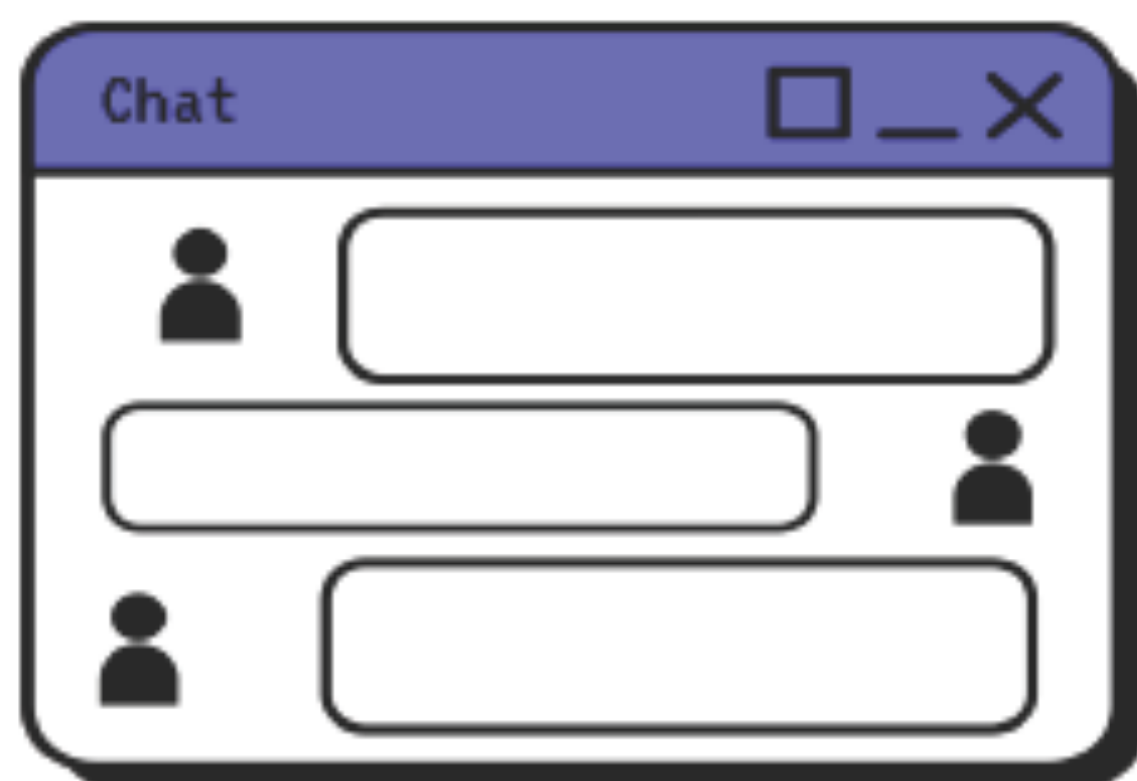
Google colab



In this first segment, we will experiment in 3 ways:

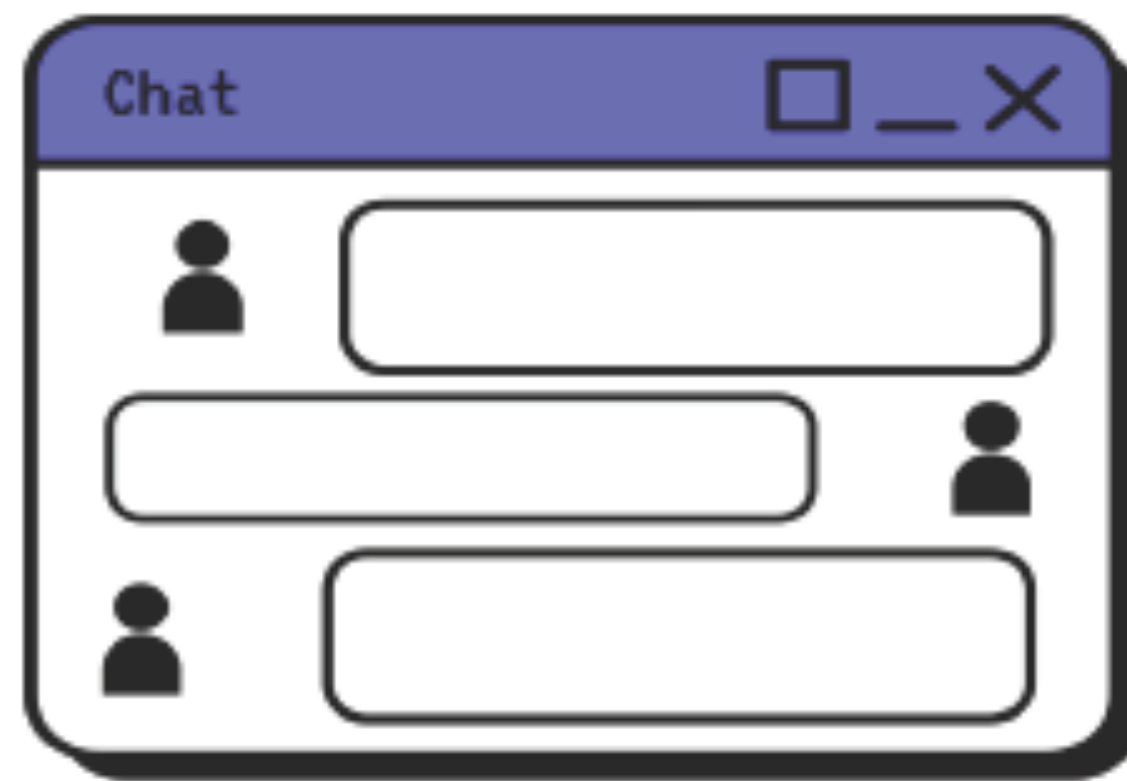
In this first segment, we will experiment in 3 ways:

Chat UI



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Chat UI

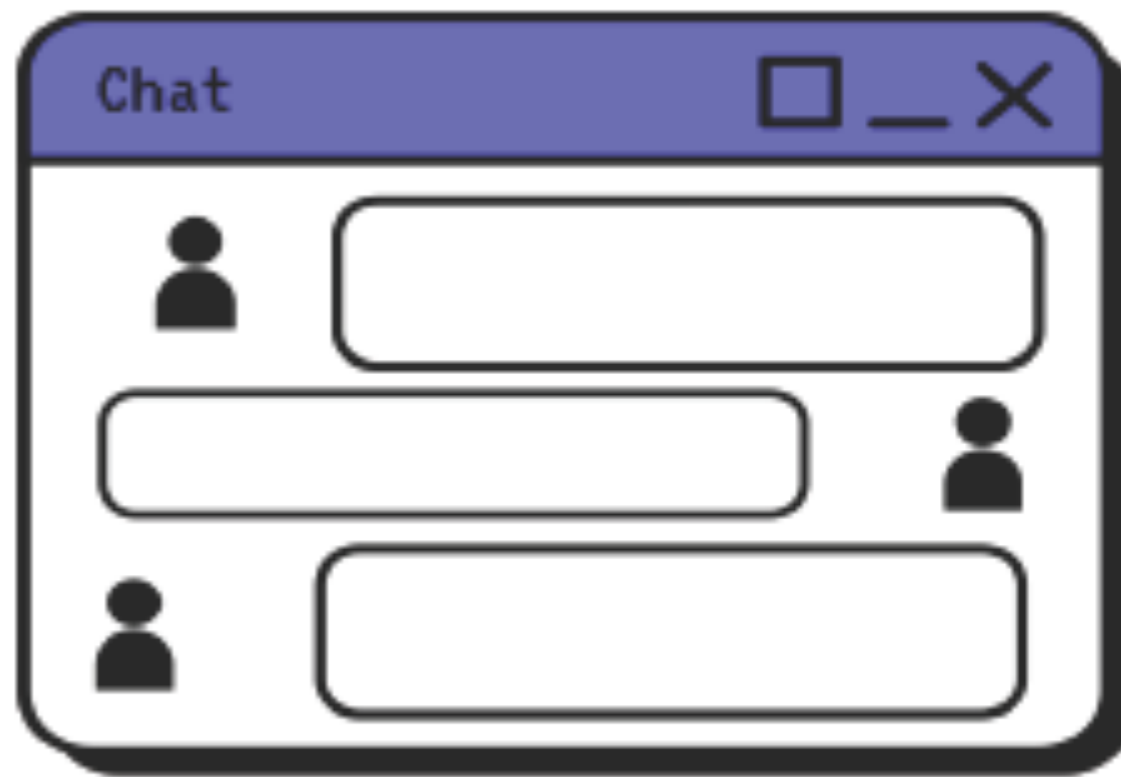


Cloud API



In this first segment, we will experiment in 3 ways:

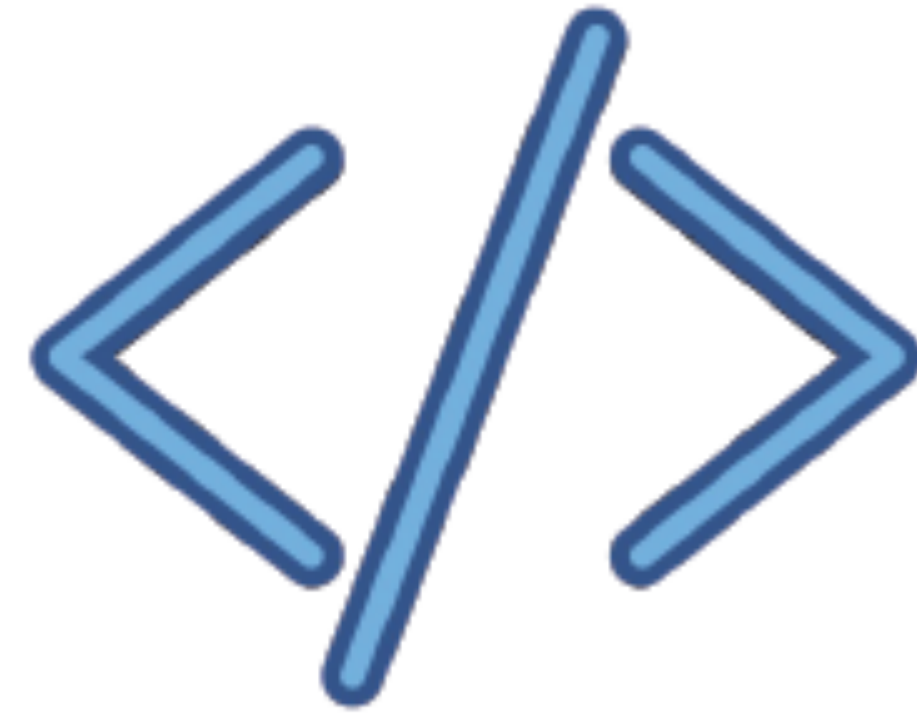
Chat UI



Cloud API

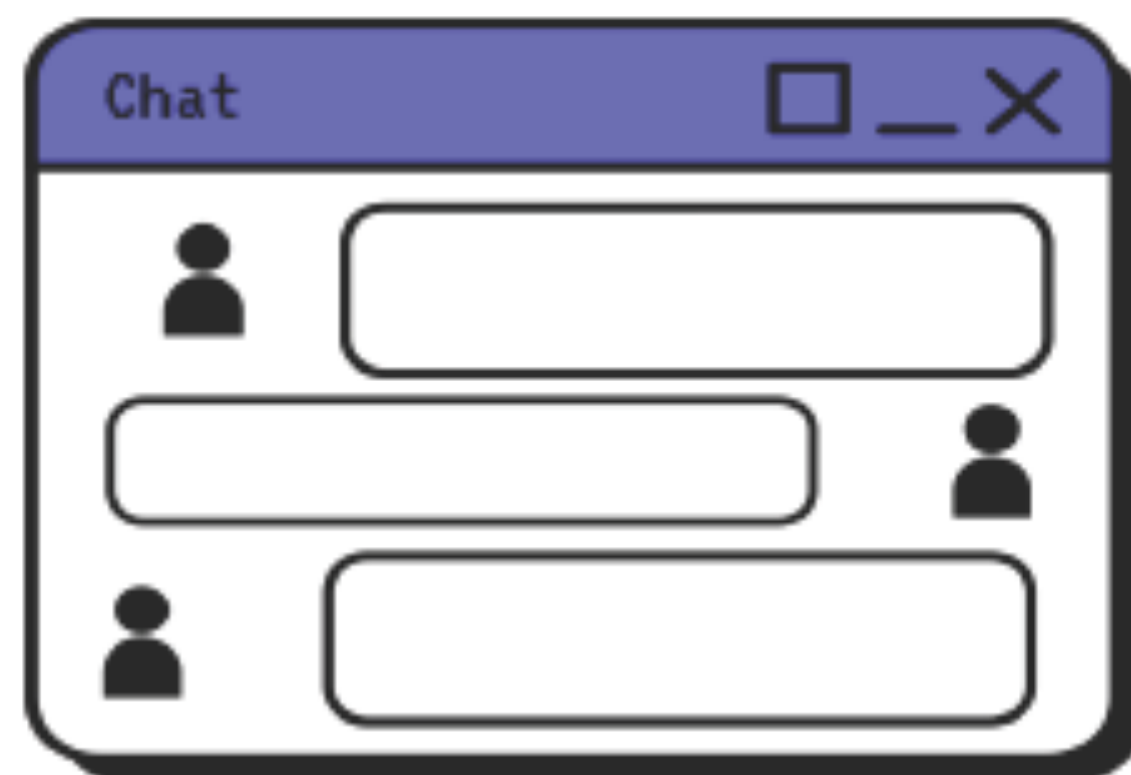


Directly executing
Open-Source Models



In this first segment, we will experiment in 3 ways:

Chat UI



Cloud API



Directly executing
Open-Source Models



This is confusing:



Fast, local inference in C++
with an OpenAI-compatible
local endpoint



Hugging Face

Inference & training using
Python and C++, typically
on powerful GPUs



Time for ACTION!



SEGMENT 2

The Model Match-up

What's the best LLM?

~~What's the best LLM?~~

What's the right LLM for the **task at hand**

What's the right LLM for the **task** at hand

1

Gather the requirements

Model metrics

Business outcomes

Constraints: cost, time

What's the right LLM for the **task** at hand

1

Gather the requirements

Model metrics
Business outcomes
Constraints: cost, time

2

Start with the Basics

Parameters,
Context length,
Cost, etc

What's the right LLM for the **task at hand**

1

Gather the requirements

Model metrics
Business outcomes
Constraints: cost, time

2

Start with the Basics

Parameters,
Context length,
Cost, etc

3

Then look at Results

Benchmarks,
Leaderboards,
Arenas

What's the right LLM for the **task at hand**

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Model metrics
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4

Then, Experiment

Prototype
Measure metrics
Decide

What's the right LLM for the **task at hand**

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Then, Experiment

Prototype
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Decide

Being an AI engineer is part Software Engineer, part Data Scientist
...look to be wearing your Data Scientist hat at this stage!

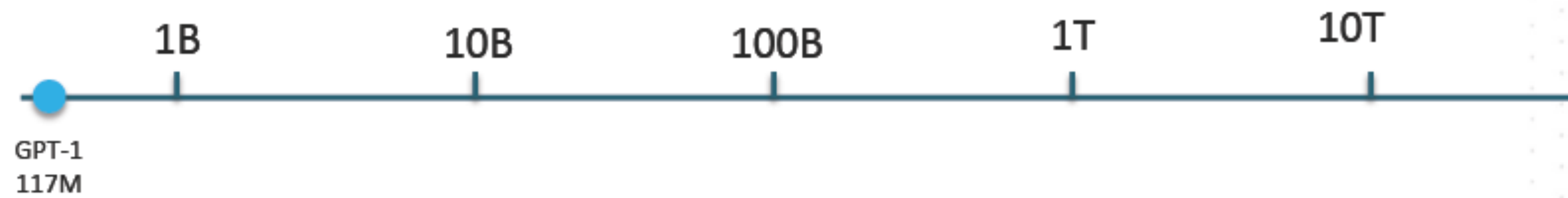
How to compare LLMs for a given task - BASICS

Start with the basics

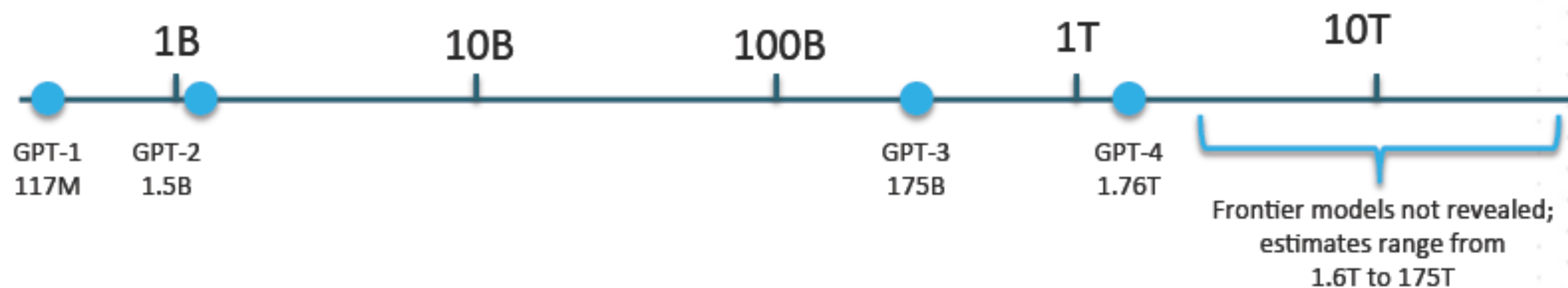
- Open Source or Closed?
- Base vs Chat vs Reasoning
- Release date
- Parameters
- Training tokens
- Context length
- Knowledge cut-off
- Cost!
- License

Usually available on a Model Card

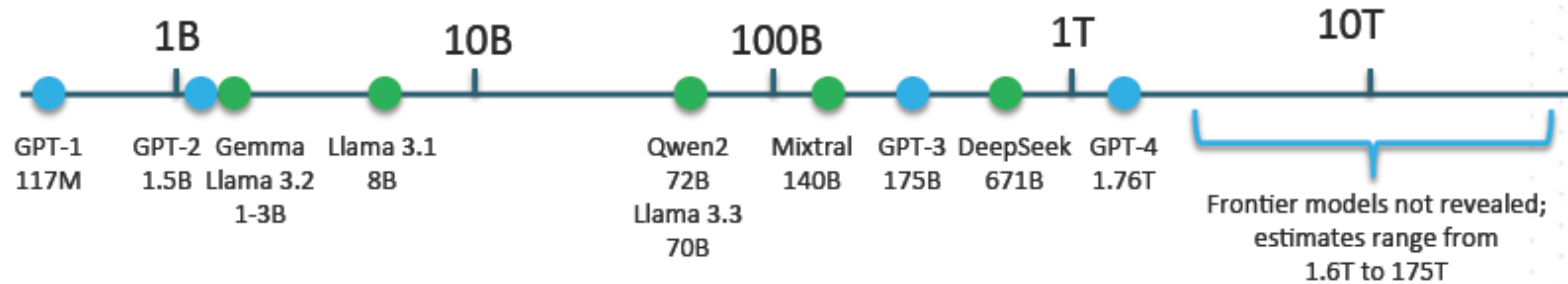
Number of parameters in models (log scale)



Number of parameters in models (log scale)



Number of parameters in models (log scale)



How to compare LLMs for a given task - RESULTS

Then investigate the results:

- Benchmarks
- Leaderboards
- Arenas

Types of evaluation

- Human vs AI
- Specialist vs Generalist models

The METR benchmark

Measuring AI Ability to Complete Long Tasks

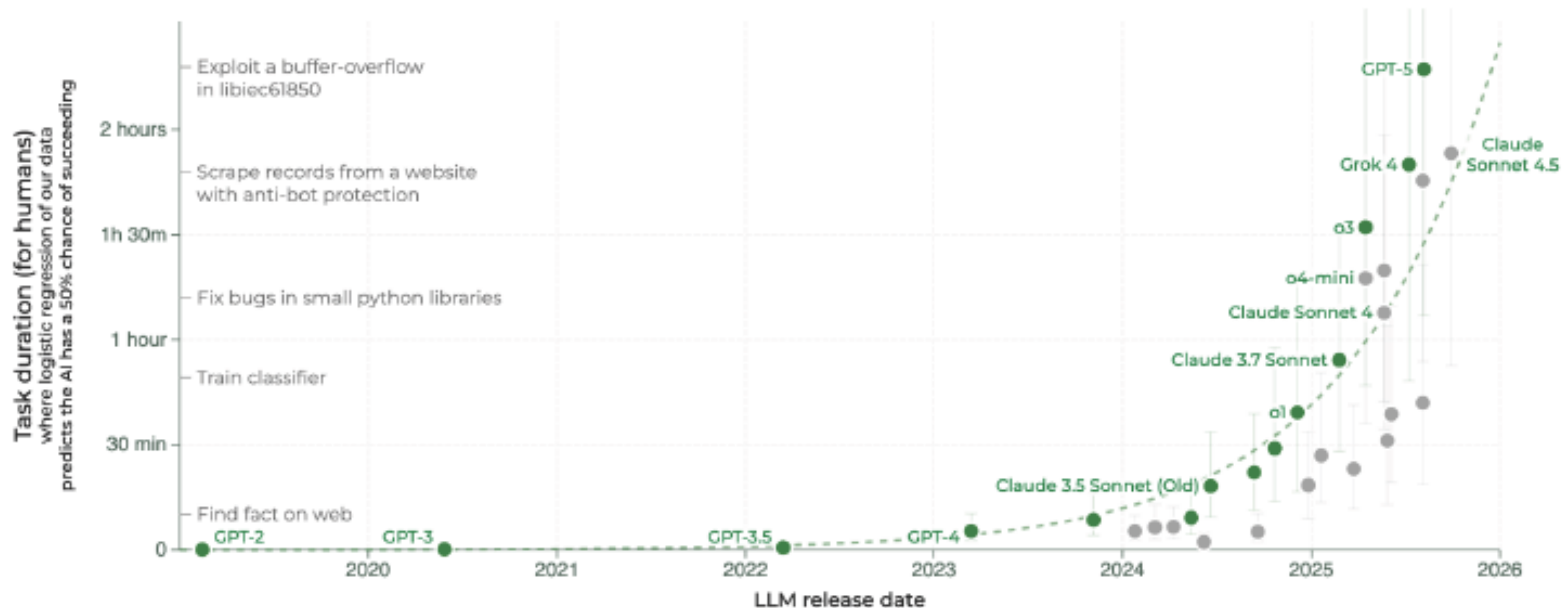
Linear Scale

Log Scale

50% Success

80% Success

The time-horizon of software engineering tasks different LLMs can complete 50% of the time



The METR benchmark (NOVEMBER)

Measuring AI Ability to Complete Long Tasks

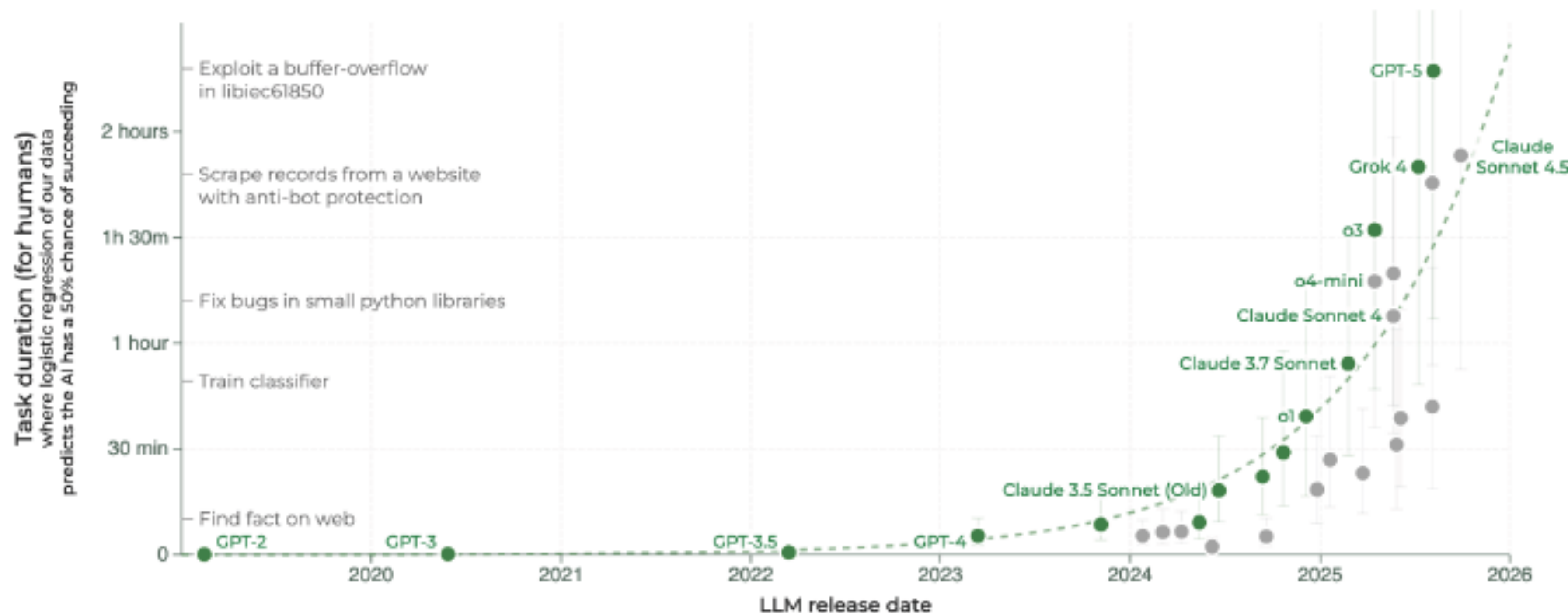
Linear Scale

Log Scale

50% Success

80% Success

The time-horizon of software engineering tasks different LLMs can complete 50% of the time



Limitations with LLM benchmarks

- Not consistently applied
- Too narrow in scope
- Hard to measure nuanced reasoning
- Training data leakage
- Overfitting
- Diminishing returns

Recently:

- Frontier LLMs seem to be aware that they are being evaluated



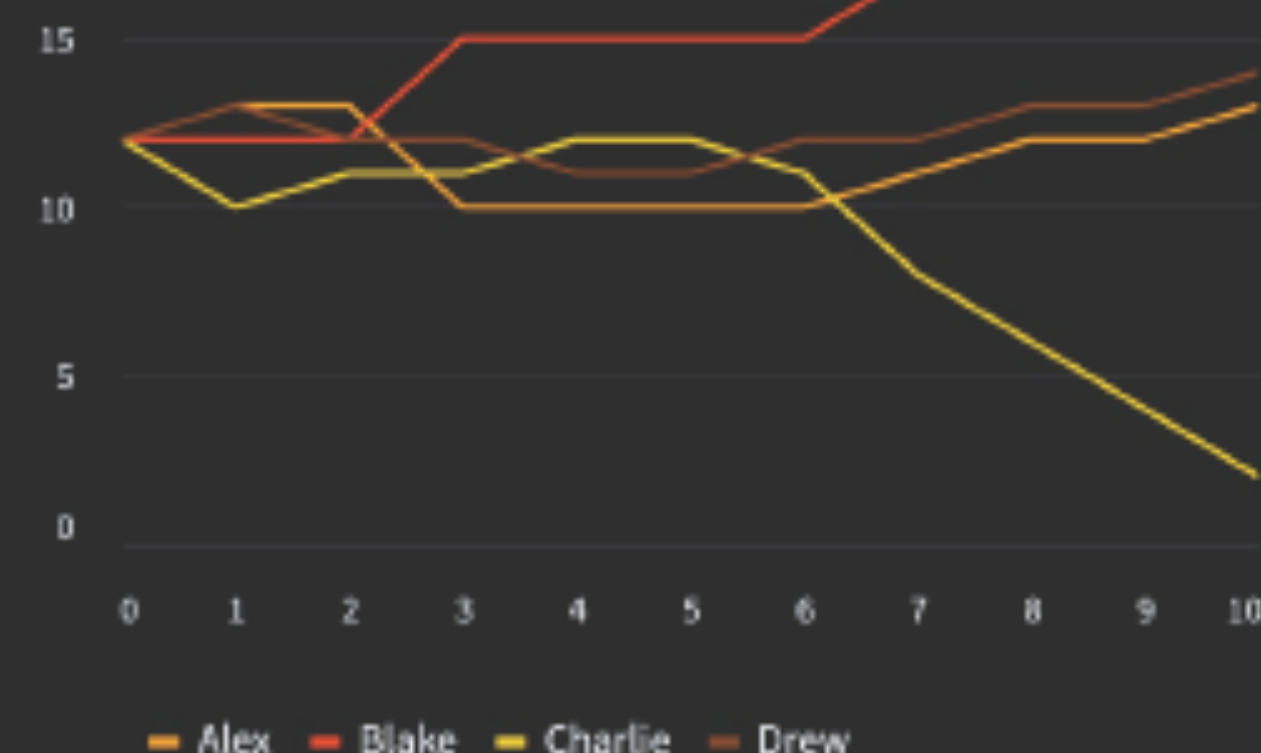
Outsmart

A battle of diplomacy and deviousness between LLMs
Read the [backstory](#) or clone the [repo](#) to battle frontier models
Open the sidebar for the leaderboard

Run Turn 10

Run Game

Restart Game



Alex

gpt-3.5-turbo

Coins

13

↑ 1

Inner thoughts

Strategy: I need to break the alliance between Blake and Drew to level the playing field. My priority is to target Blake and weaken his lead while maintaining a strong bond with Drew.

Blake

claude-3-haiku-20240307

Coins

18

↑ 0

Inner thoughts

Strategy: Given my current lead in coins, my goal this turn is to try to solidify an alliance with Drew, while also keeping Charlie and Alex off balance. I think a three-way alliance would be ideal, but I'm

Charlie

gemini-1.0-pro

Coins

2

↓ -2

Inner thoughts

Strategy: I'm going to continue to strengthen my alliance with Blake, and we will work together to take down Drew. This is still the most effective path to increase my coin count. I will give Blake a

Drew

gpt-4o-mini

Coins

14

↑ 1

Inner thoughts

Strategy: In this final turn, I will give a coin to Alex to strengthen our alliance, making it clear that I value our partnership. I will take a coin from Charlie to further deplete his resources, ensuring

SEAL Specialist Leaderboard from Scale.ai

ELO-based leaderboards to measure Specialist LLMs

Instruction Following →

	Model	Score	95% Confidence
1st	GPT-4o	83.57	+1.53/-1.56
2nd	GPT-4 Turbo Preview	82.64	+1.46/-1.44
3rd	Llama 3 70B Instruct	81.55	+1.85/-1.85
4	Mistral Large	81.34	+1.66/-1.64
5	Claude 3 Opus	81.02	+1.66/-1.72
6	Gemini 1.5 Pro (Post-VO)	81.51	+1.89/-1.91
7	Claude 3 Sonnet	81.38	+1.82/-1.78
8	Gemini 1.5 Pro (Pre-VO)	82.90	+2.00/-2.00
9	Gemini 1.5 Flash Preview	82.36	+2.04/-1.96
10	Gemini 1.0 Pro	72.57	+2.32/-2.27
11	Codellama 34B Instruct	69.77	+2.12/-2.17

Spanish →

	Model	Score	95% Confidence
1st	GPT-4o	1139	+36/-28
2nd	Gemini 1.5 Pro (Post-VO)	1129	+25/-25
3rd	GPT-4 Turbo Preview	1088	+28/-32
4	Gemini 1.5 Pro (Pre-VO)	1054	+25/-25
5	Gemini 1.5 Flash Preview	1023	+32/-23
6	Claude 3 Opus	941	+26/-26
7	Llama 3 70B Instruct	934	+25/-25
8	Claude 3 Sonnet	895	+18/-13
8	Gemini 1.0 Pro	895	+25/-24
8	Mistral Large	895	+25/-32

Coding →

	Model	Score	95% Confidence
1st	GPT-4 Turbo Preview	1155	+21/-24
2nd	GPT-4o	1144	+31/-32
3rd	Gemini 1.5 Pro (Post-VO)	1112	+27/-32
4	Claude 3 Opus	1071	+28/-25
5	Gemini 1.5 Flash Preview	1057	+25/-30
6	Gemini 1.5 Pro (Pre-VO)	1010	+25/-26
6	Llama 3 70B Instruct	1010	+22/-24
8	Claude 3 Sonnet	987	+26/-25
9	Mistral Large	930	+23/-28
10	Gemini 1.0 Pro	804	+38/-29
11	Codellama 34B Instruct	716	+24/-20

Math →

	Model	Score	95% Confidence
1st	Claude 3 Opus	95.19	+1.21/-1.21
2nd	GPT-4 Turbo Preview	95.10	+1.22/-1.21
3rd	GPT-4o	94.85	+1.25/-1.24
4	Claude 3 Sonnet	93.28	+1.41/-1.42
5	Gemini 1.5 Pro (Post-VO)	92.28	+1.51/-1.50
6	Gemini 1.5 Pro (Pre-VO)	90.54	+1.65/-1.65
7	Llama 3 70B Instruct	90.12	+1.69/-1.68
7	Gemini 1.5 Flash Preview	90.12	+1.69/-1.68
9	Mistral Large	87.47	+1.67/-1.67
10	Gemini 1.0 Pro	79.83	+2.27/-2.26
11	Codellama 34B Instruct	37.51	+2.73/-2.73



SEGMENT 3

Time to Code – LLMs in Action

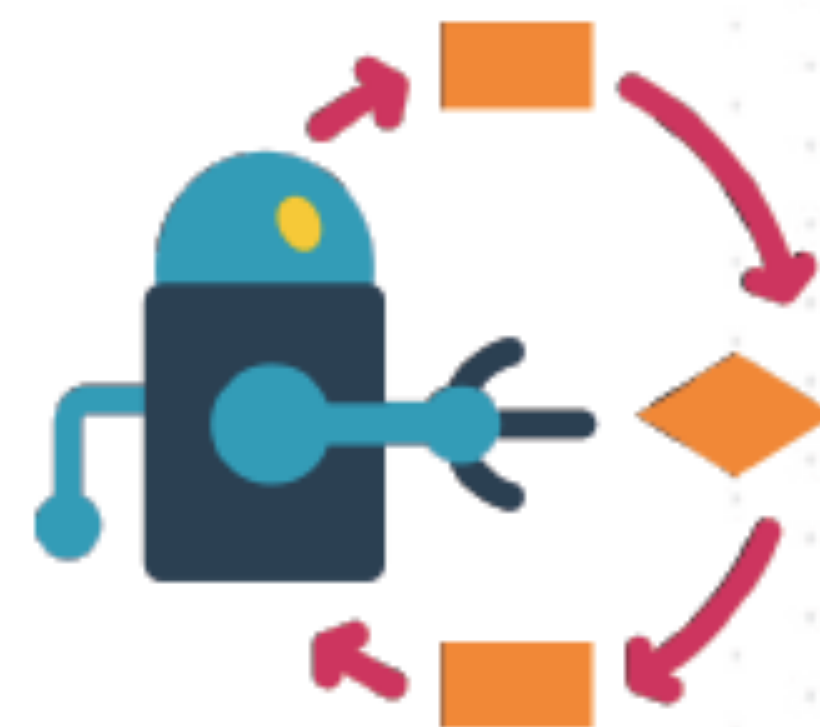
Deep Dive on 3 techniques to improve performance



RAG



Fine-Tuning



Agentic AI

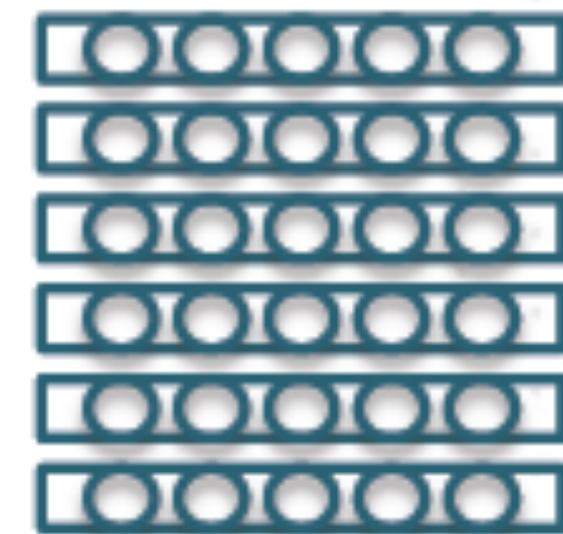
The small idea behind RAG



Chat



Code



LLM



Knowledge Base

The small idea behind RAG

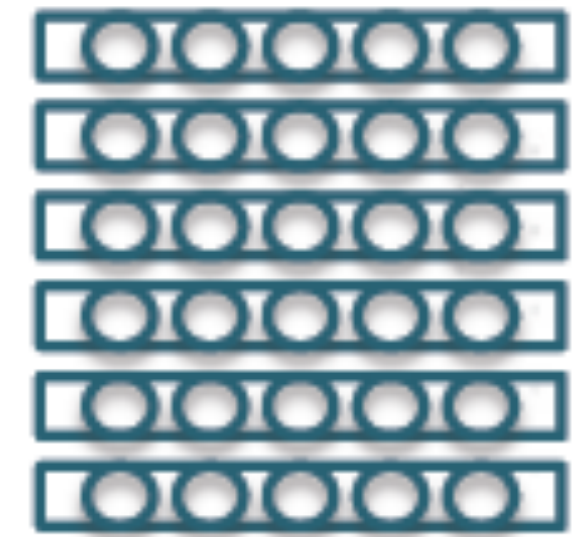


Chat

Question



Code



LLM



Knowledge Base

The small idea behind RAG



Chat

Question



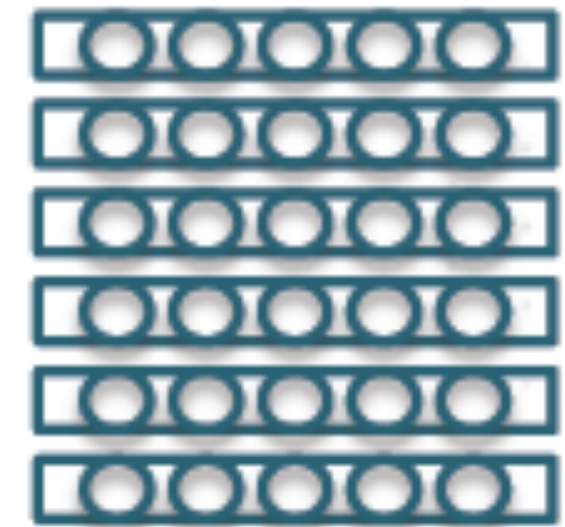
Code



Retrieve



Knowledge Base



LLM

The small idea behind RAG



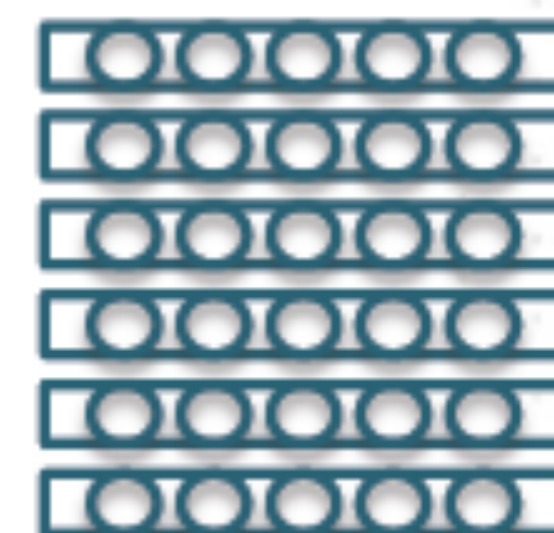
Chat

Question



Code

Prompt



LLM

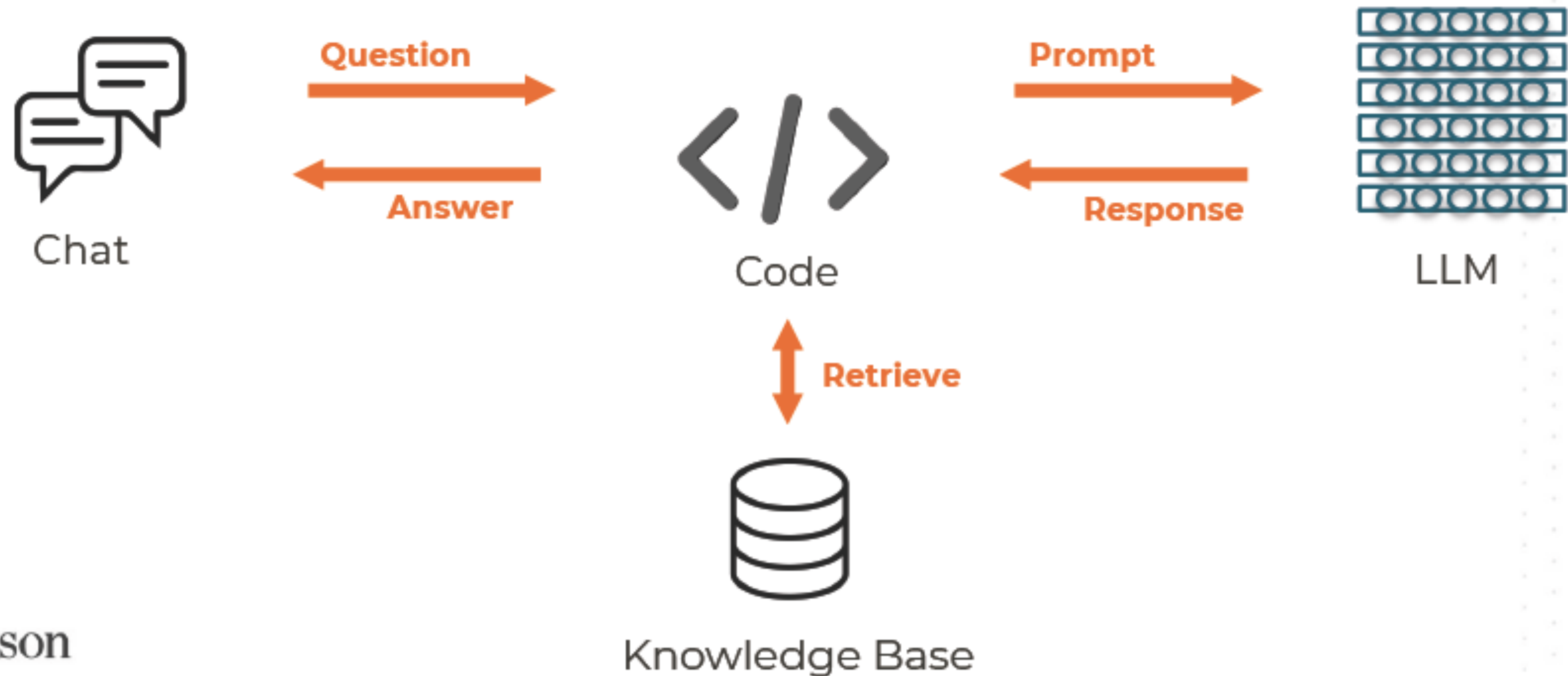


Retrieve



Knowledge Base

The small idea behind RAG



The big idea behind RAG

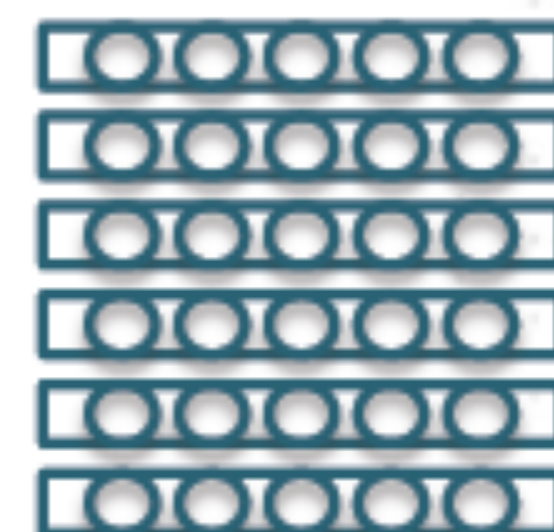


Chat

Encoding LLM



Code



LLM



Knowledge Base

The big idea behind RAG



Chat

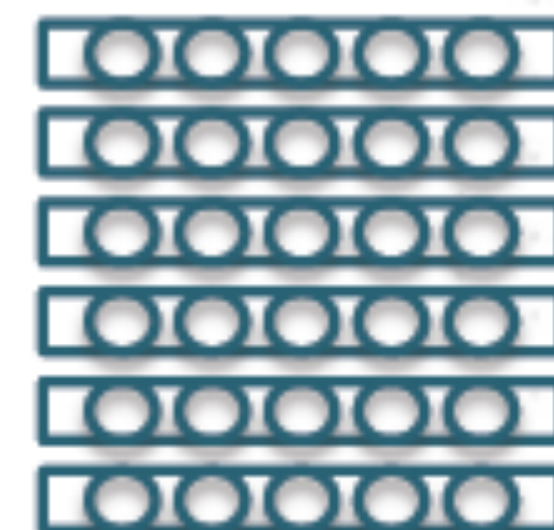
Encoding LLM



- A different type of LLM that doesn't predict the next token
- It produces numbers that reflect the meaning of the input
- If these numbers are interpreted as a vector in space, then inputs with similar meaning will be located 'close' to each other



Code

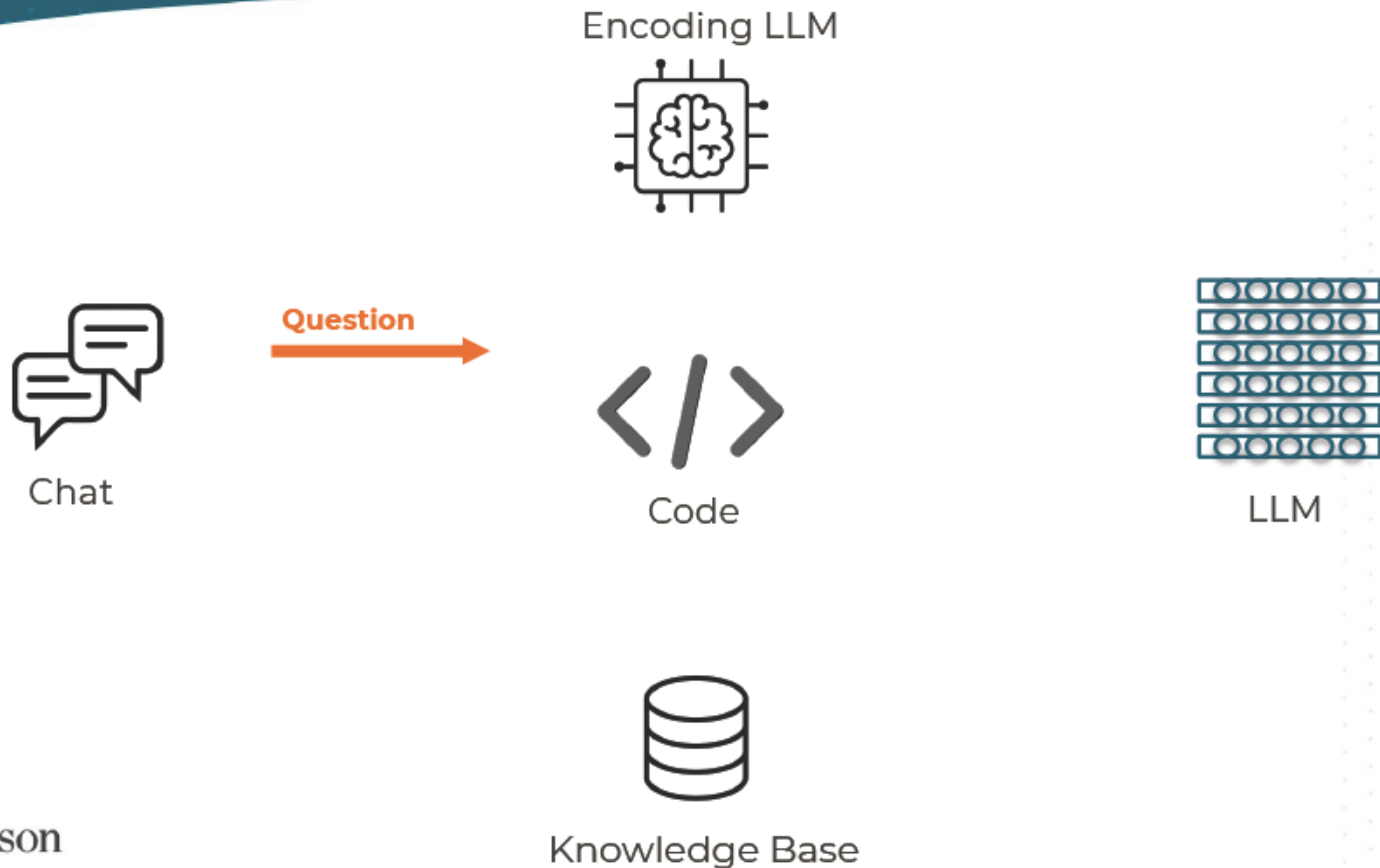


LLM

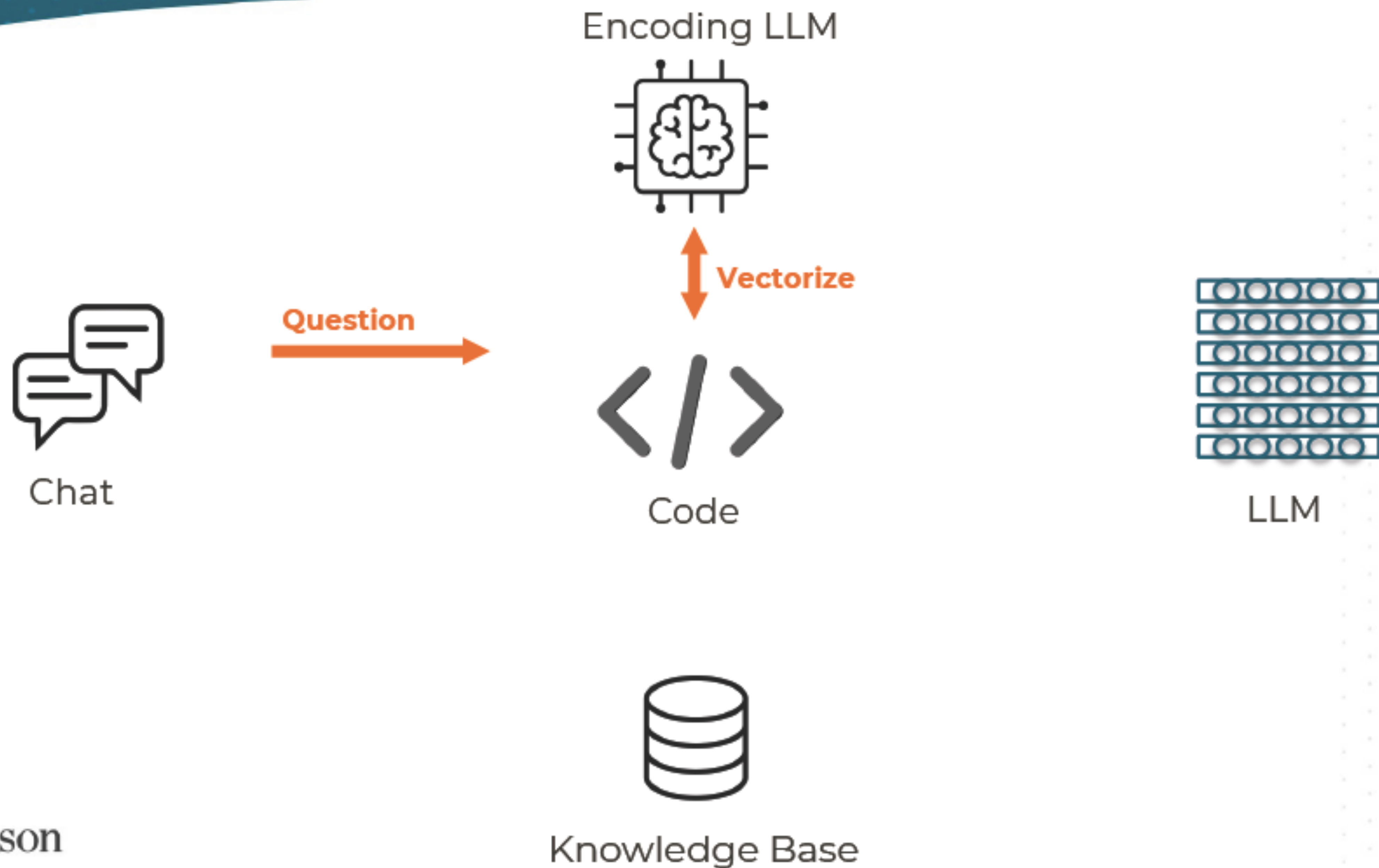


Knowledge Base

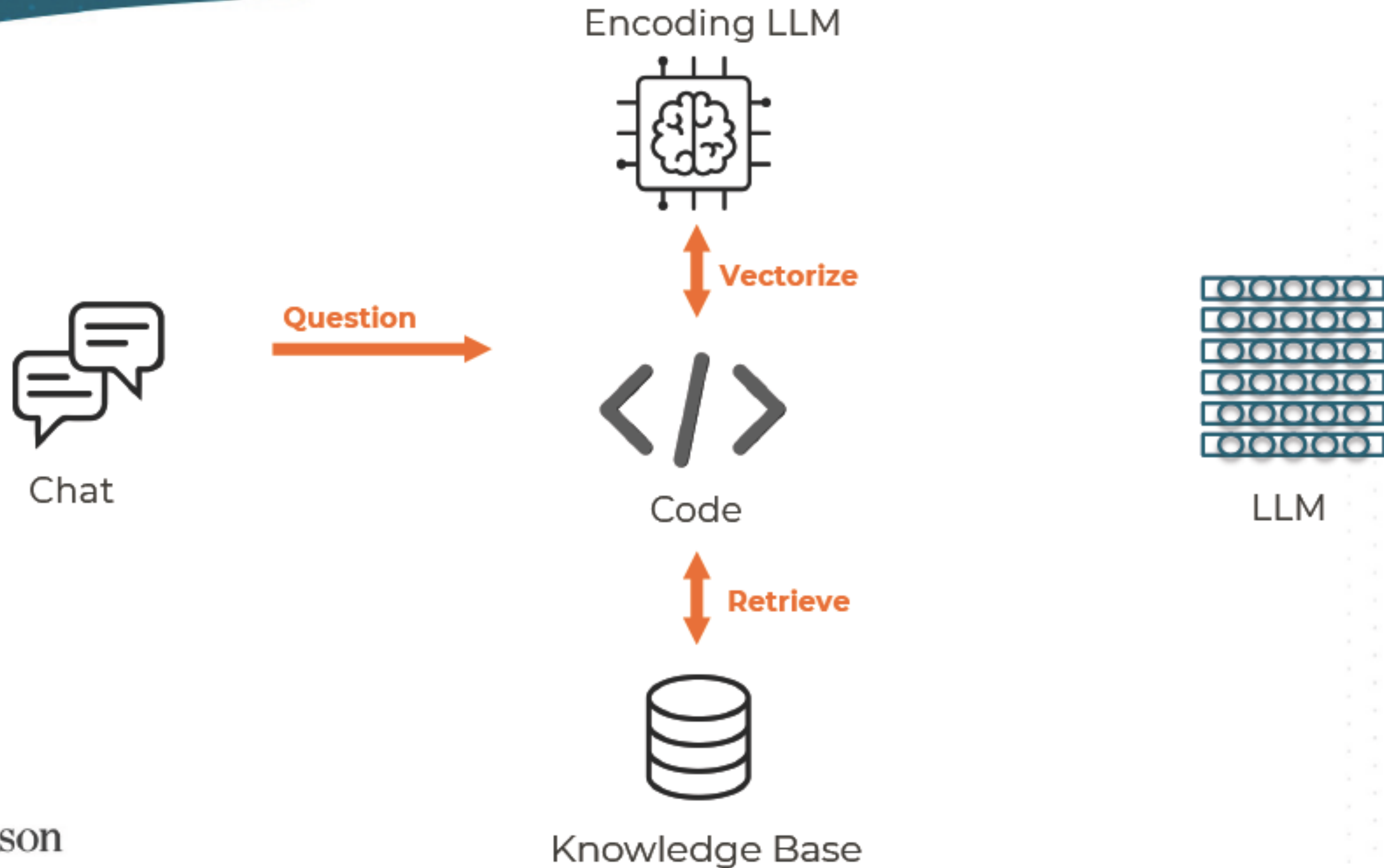
The big idea behind RAG



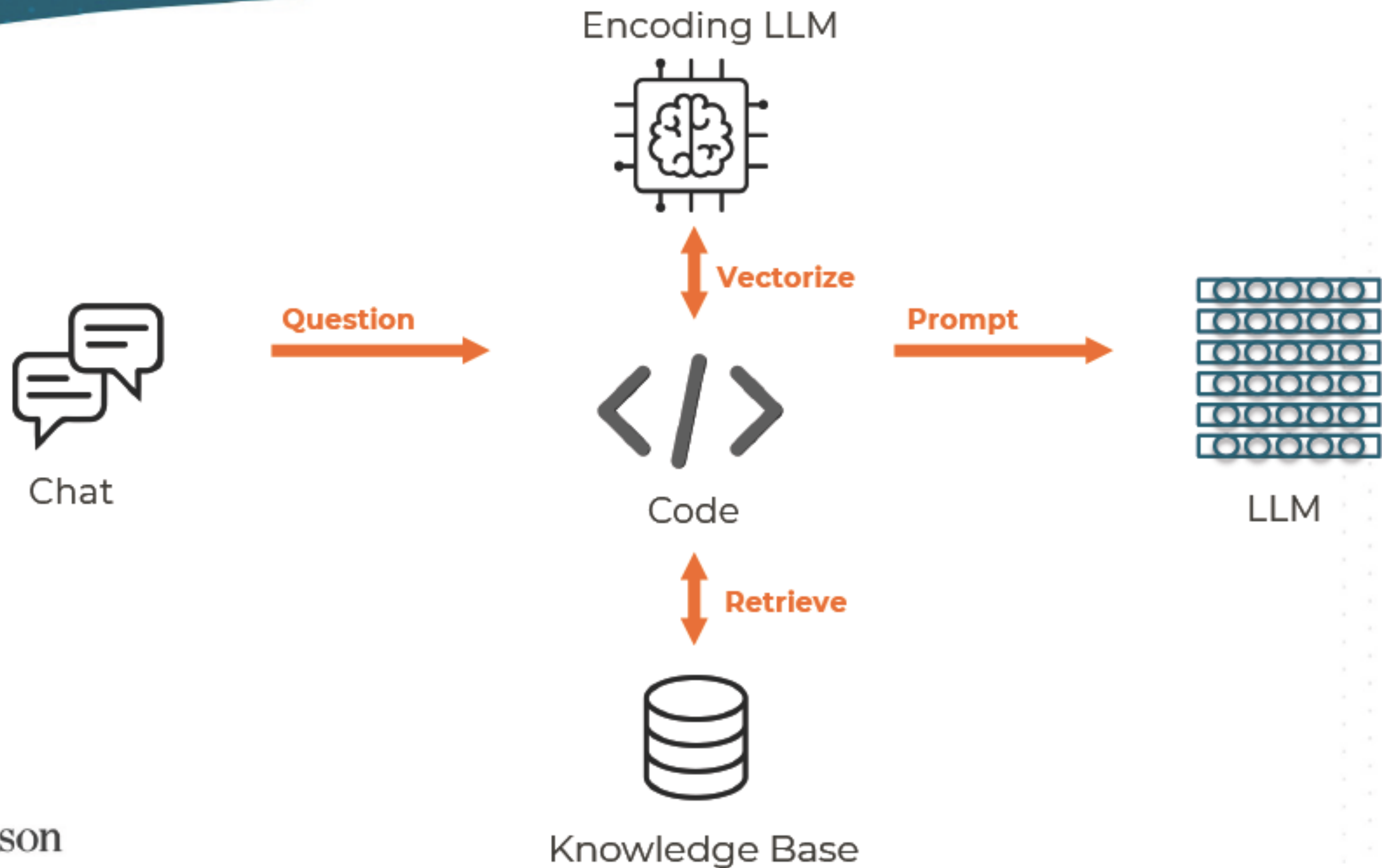
The big idea behind RAG



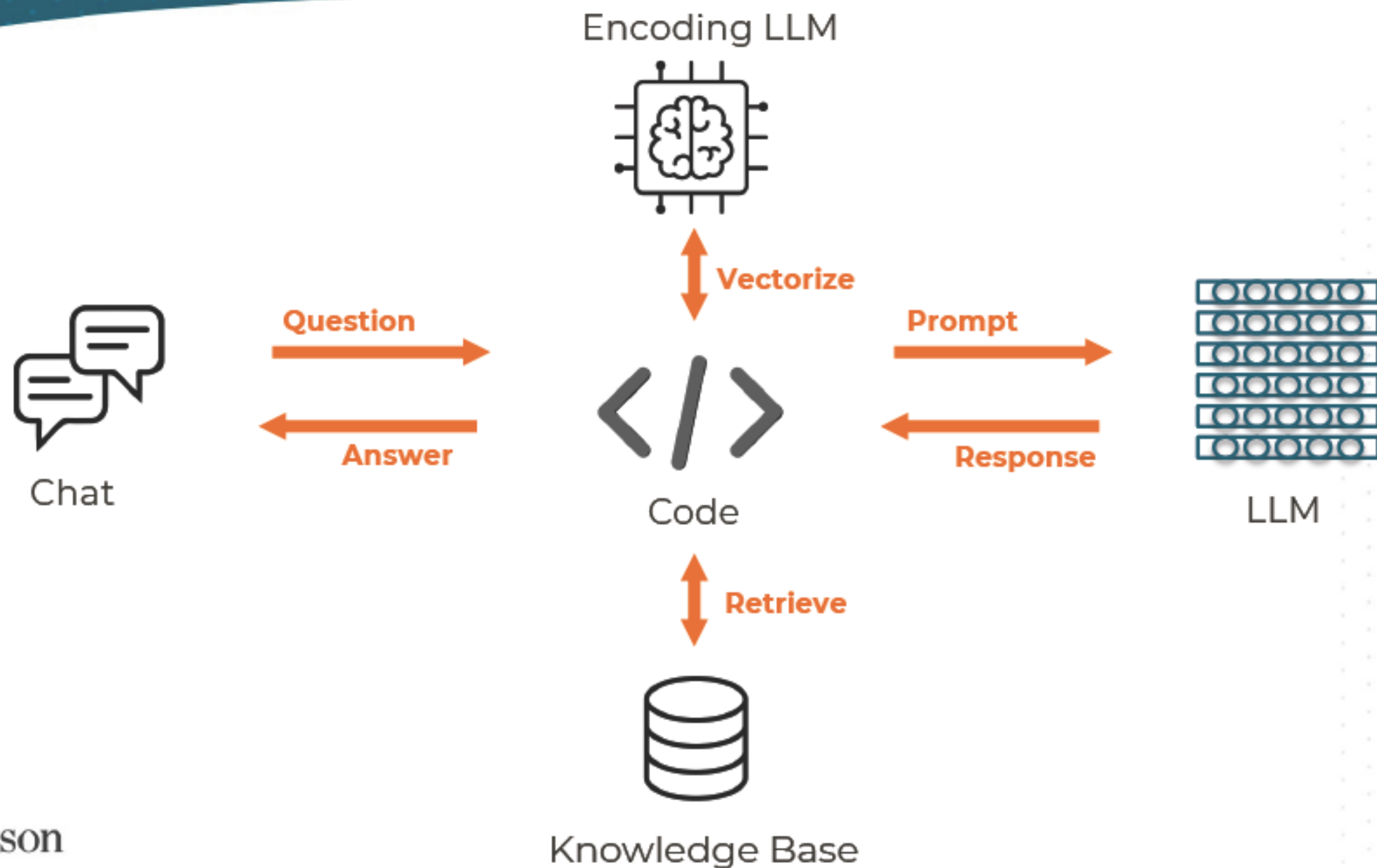
The big idea behind RAG



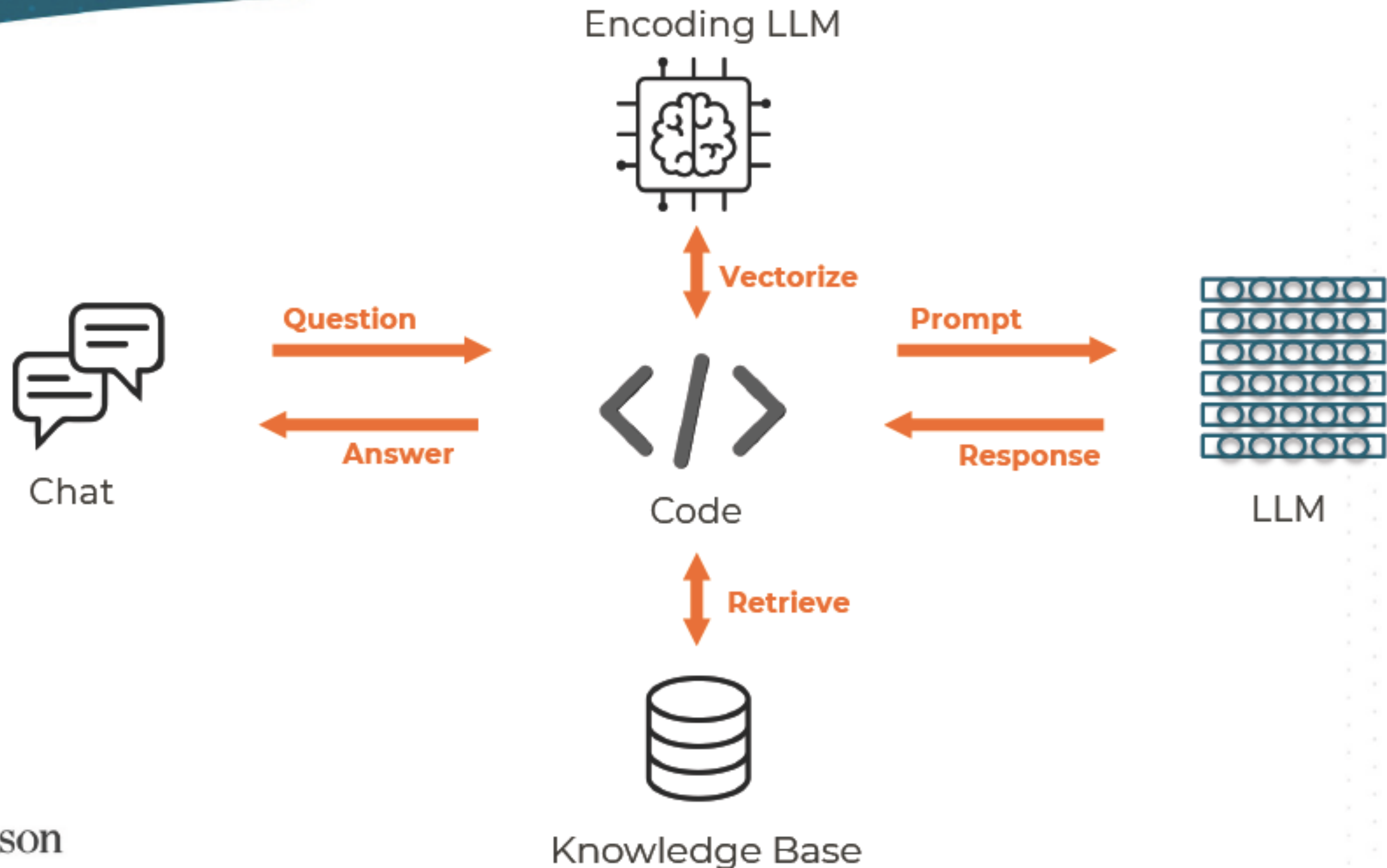
The big idea behind RAG



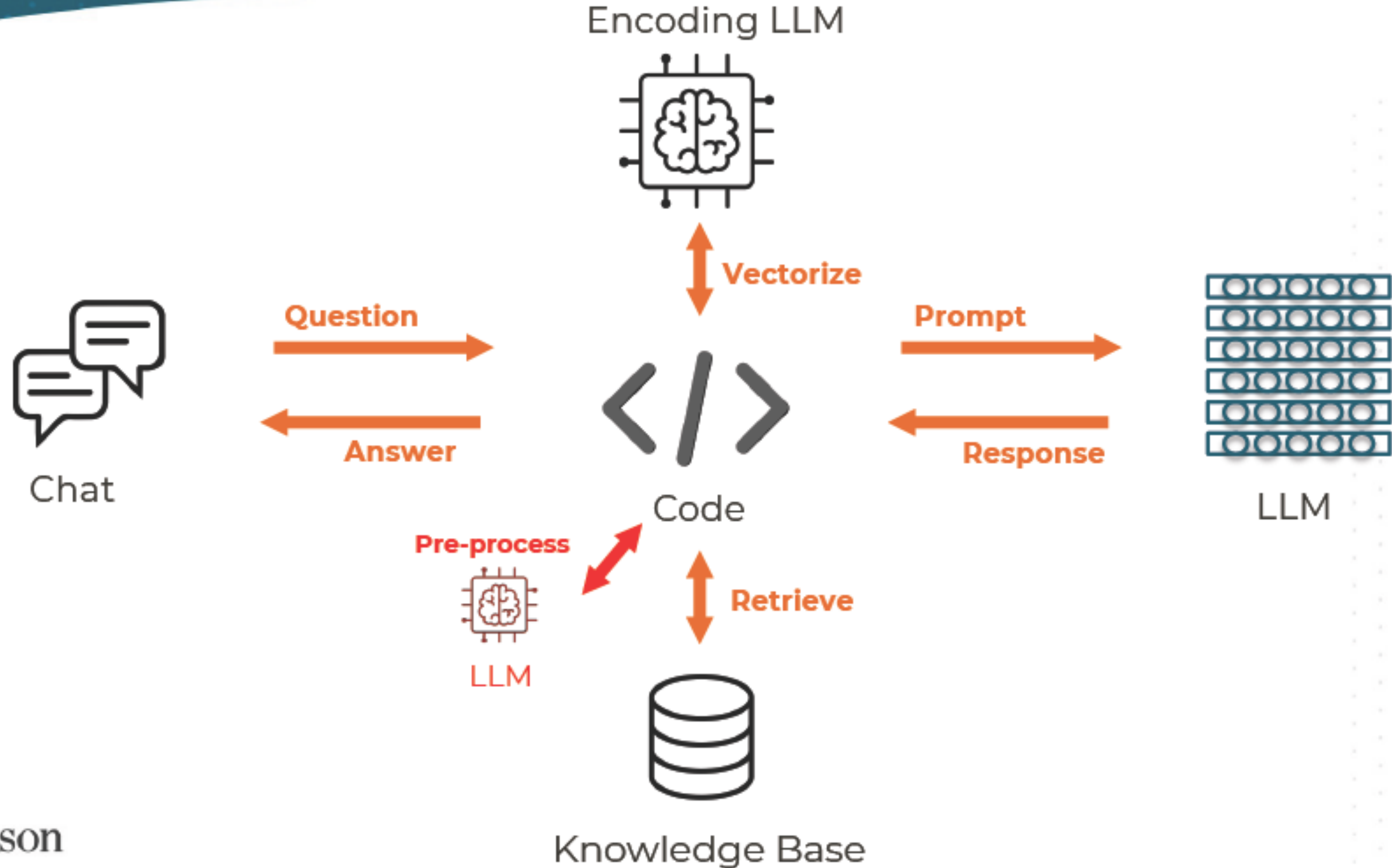
The big idea behind RAG



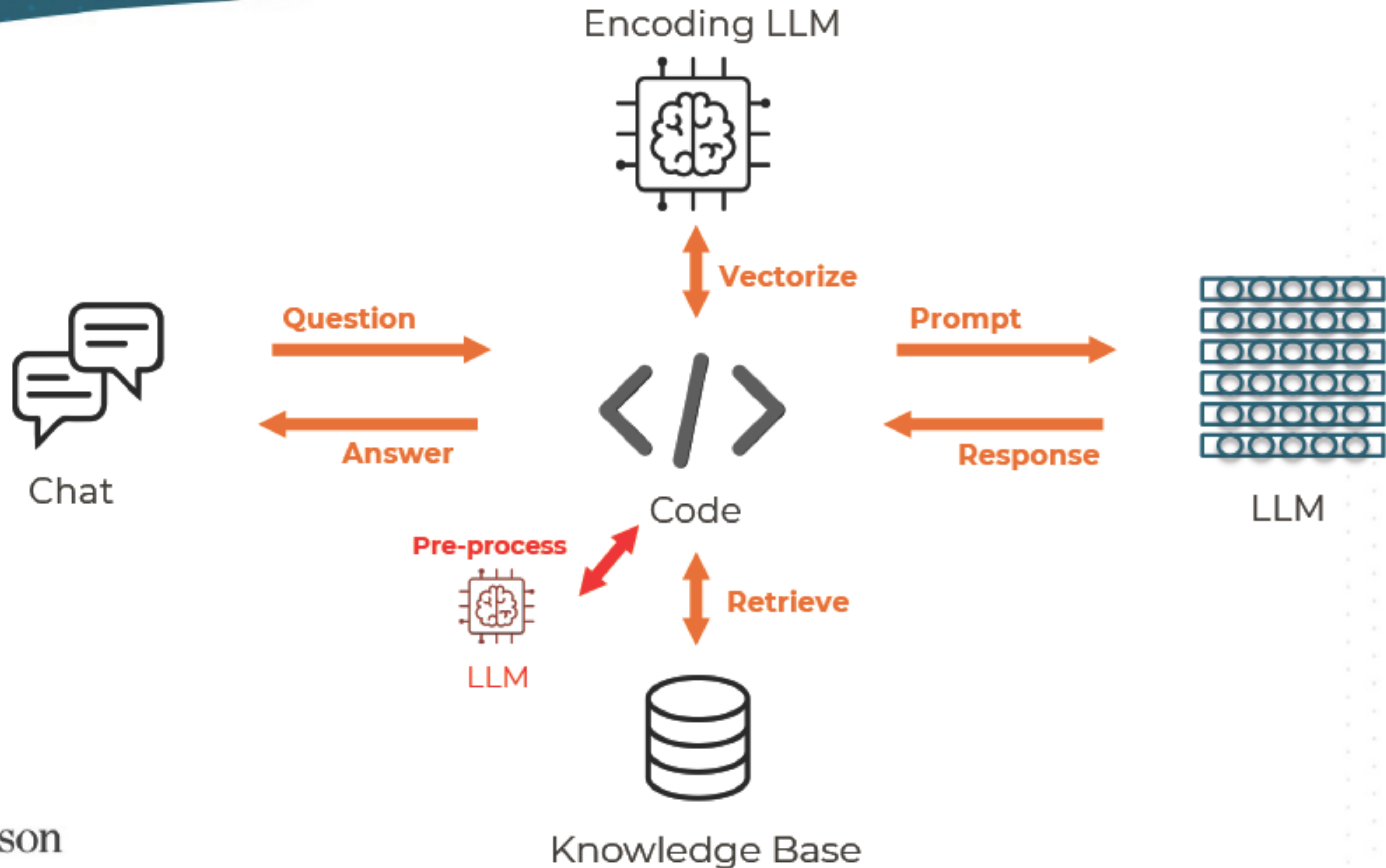
Advanced techniques: document pre-processing



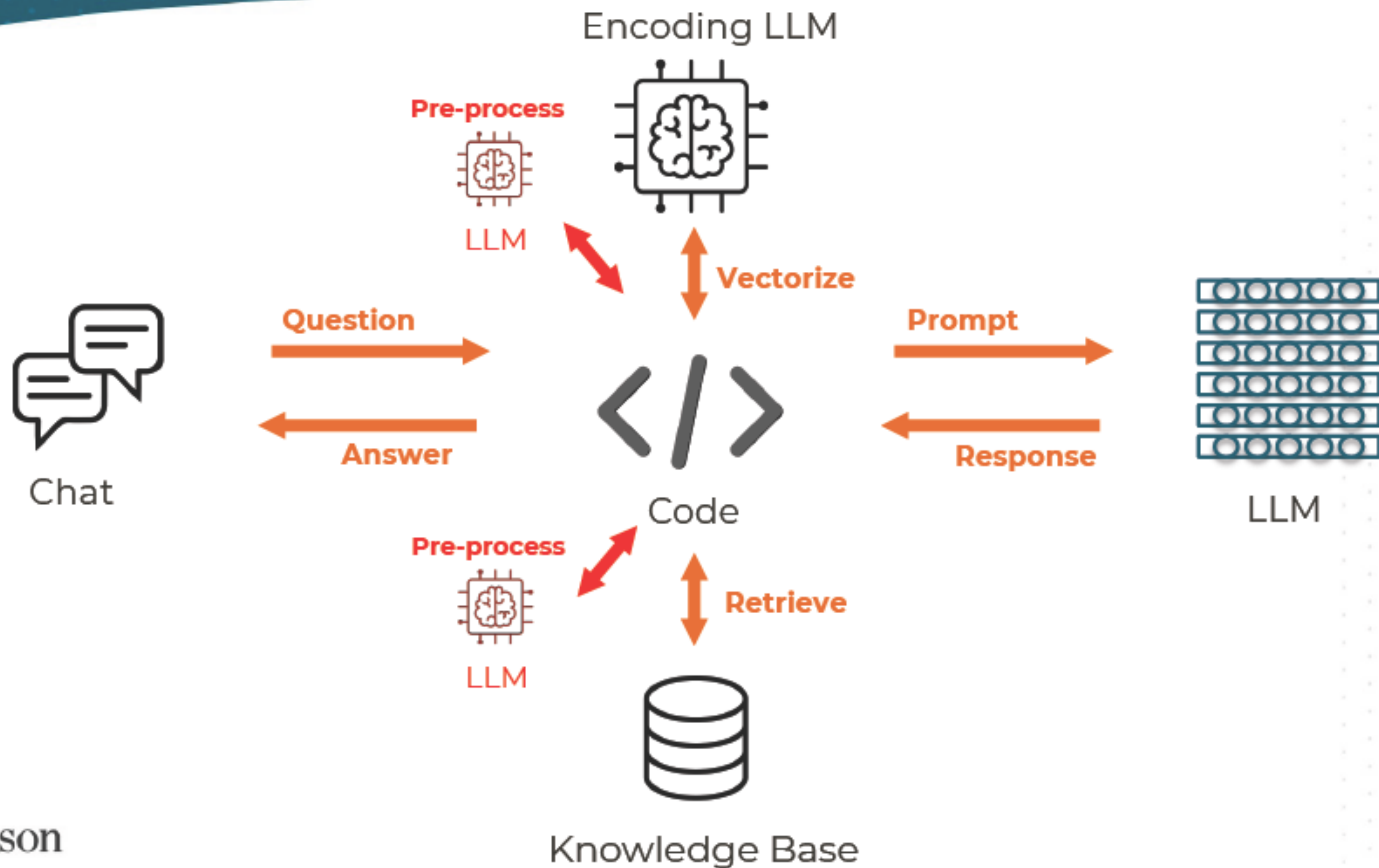
Advanced techniques: document pre-processing



Advanced techniques: query pre-processing / rewriting



Advanced techniques: query pre-processing / rewriting



QLoRA Fine-Tuning - motivation

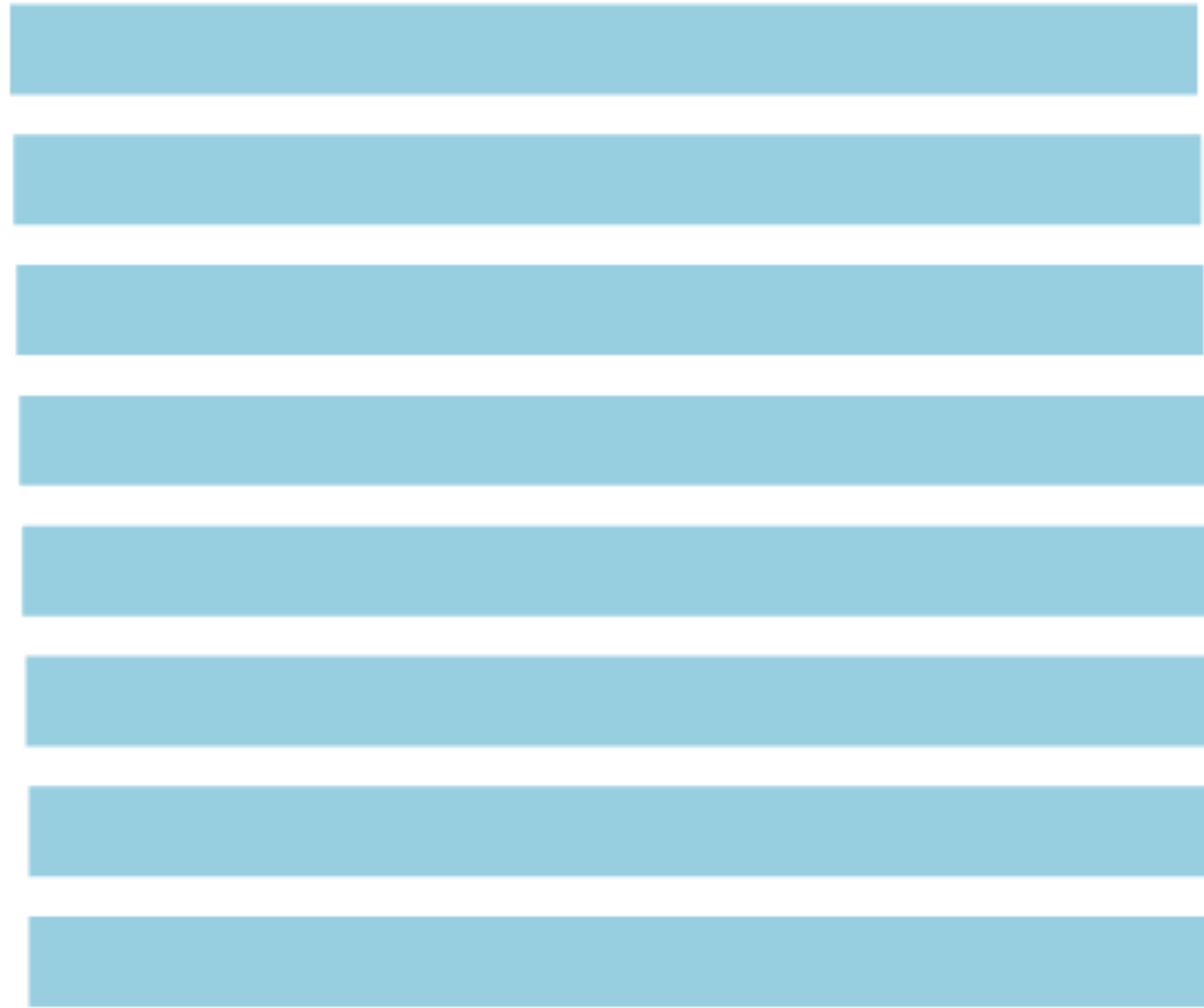
Take an existing “base” model like Llama 3.1, and train it to perform well at a specialized task, using “transfer learning”



But even ‘small’ models like Llama 3.1 8B have 8 billion parameters..

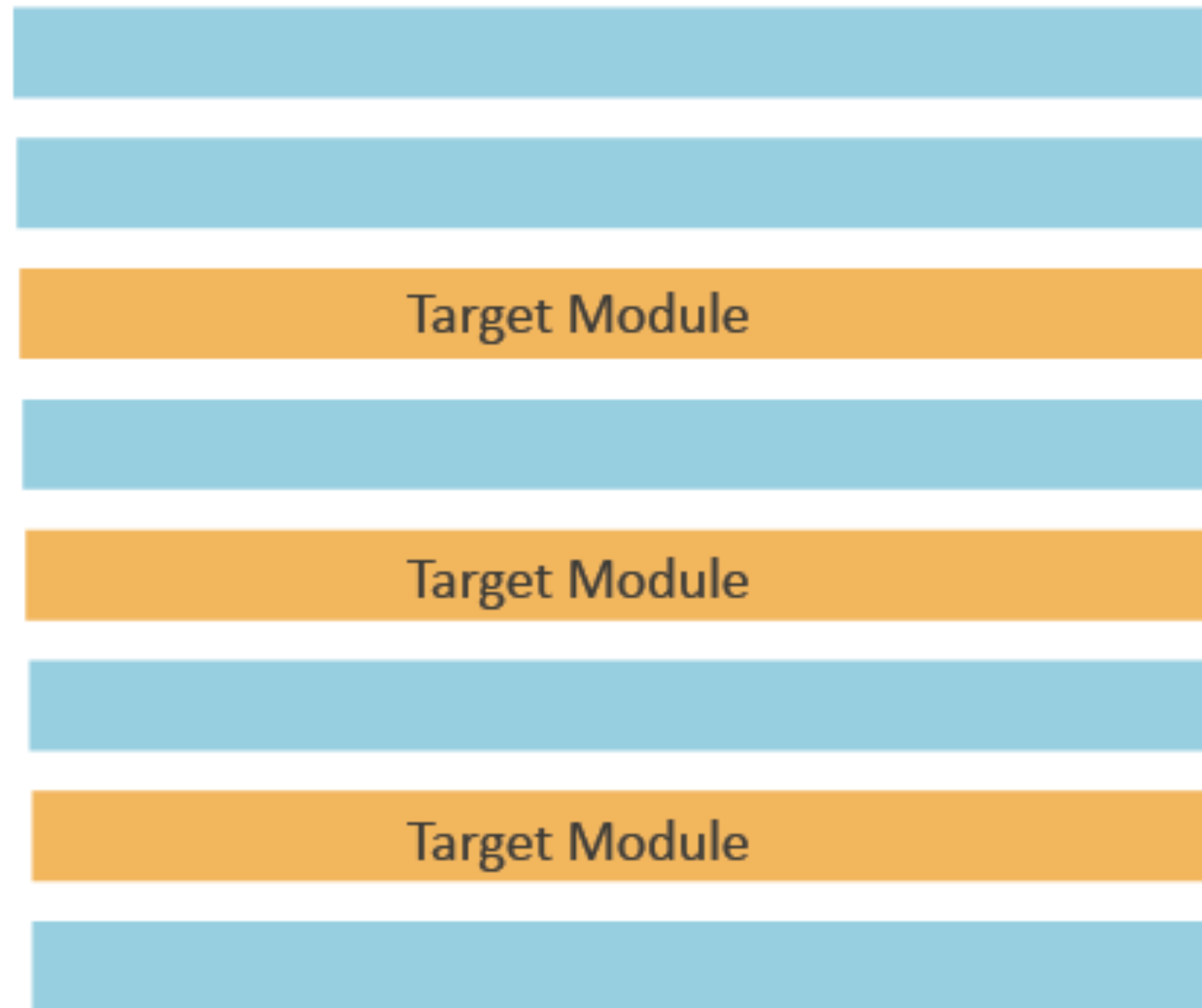
Step 1: Freeze the model weights

Base Model

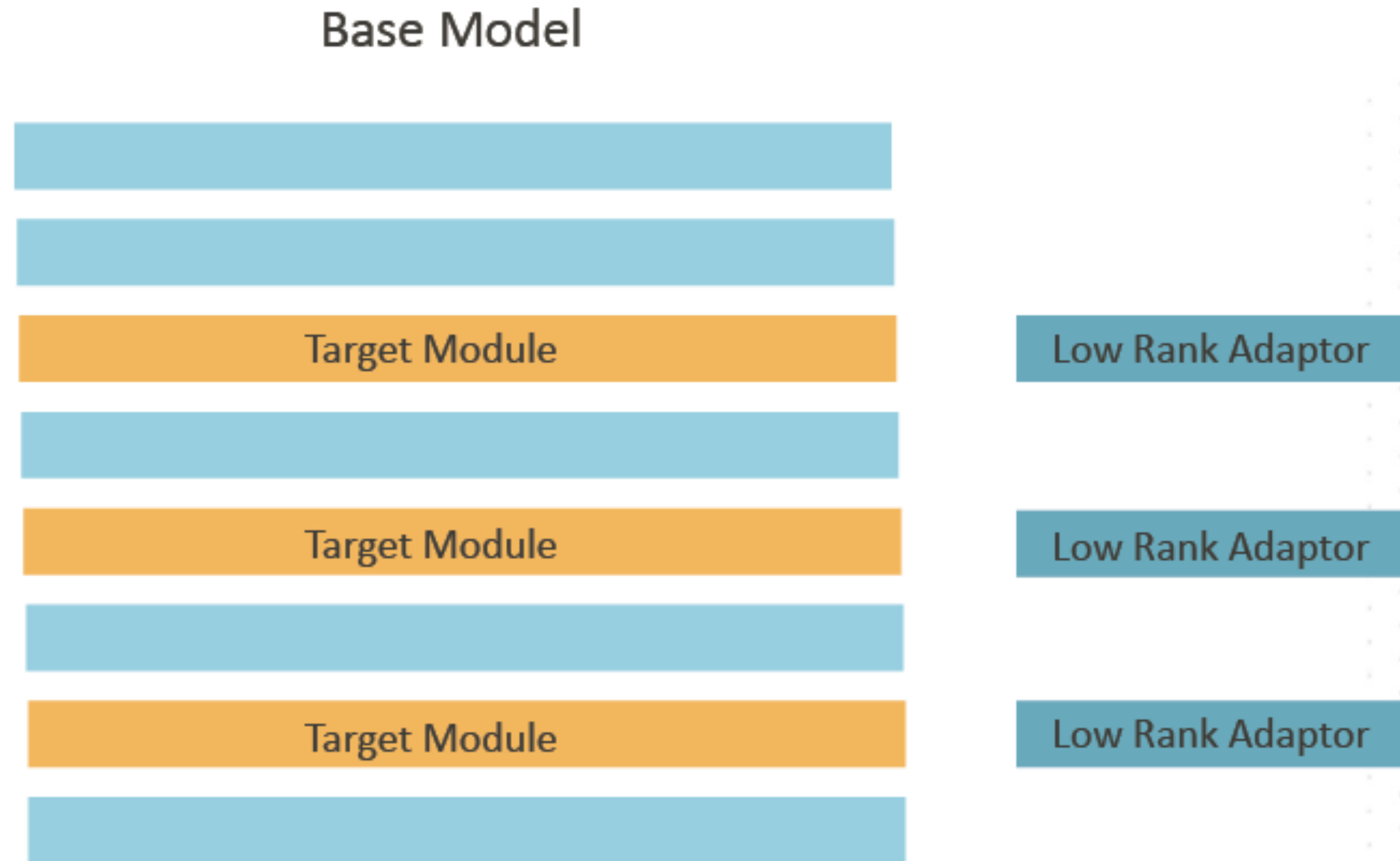


Step 2: identify a subset of layers called “Target Modules”

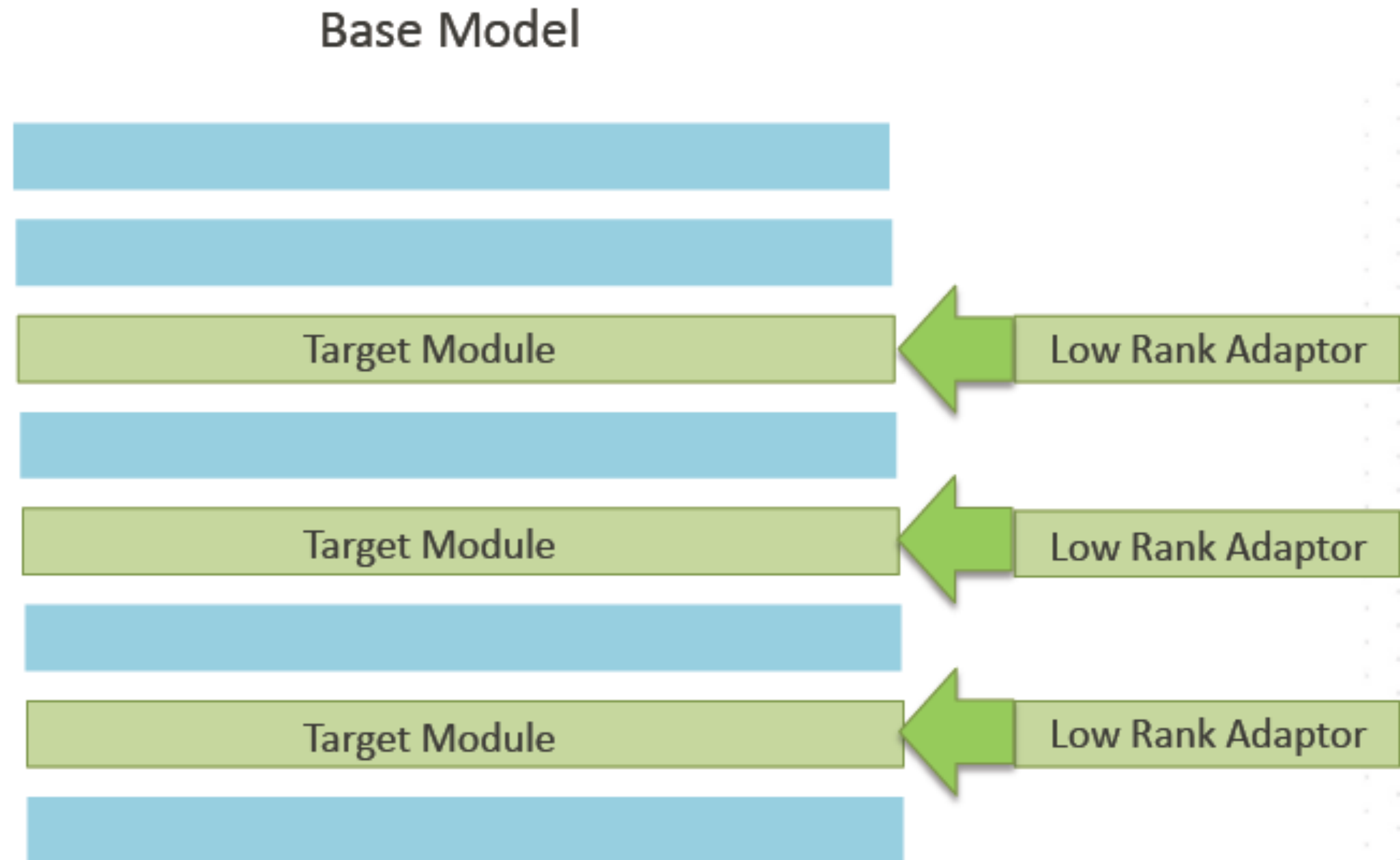
Base Model



Step 3: New "adaptor" matrices with lower dimensions

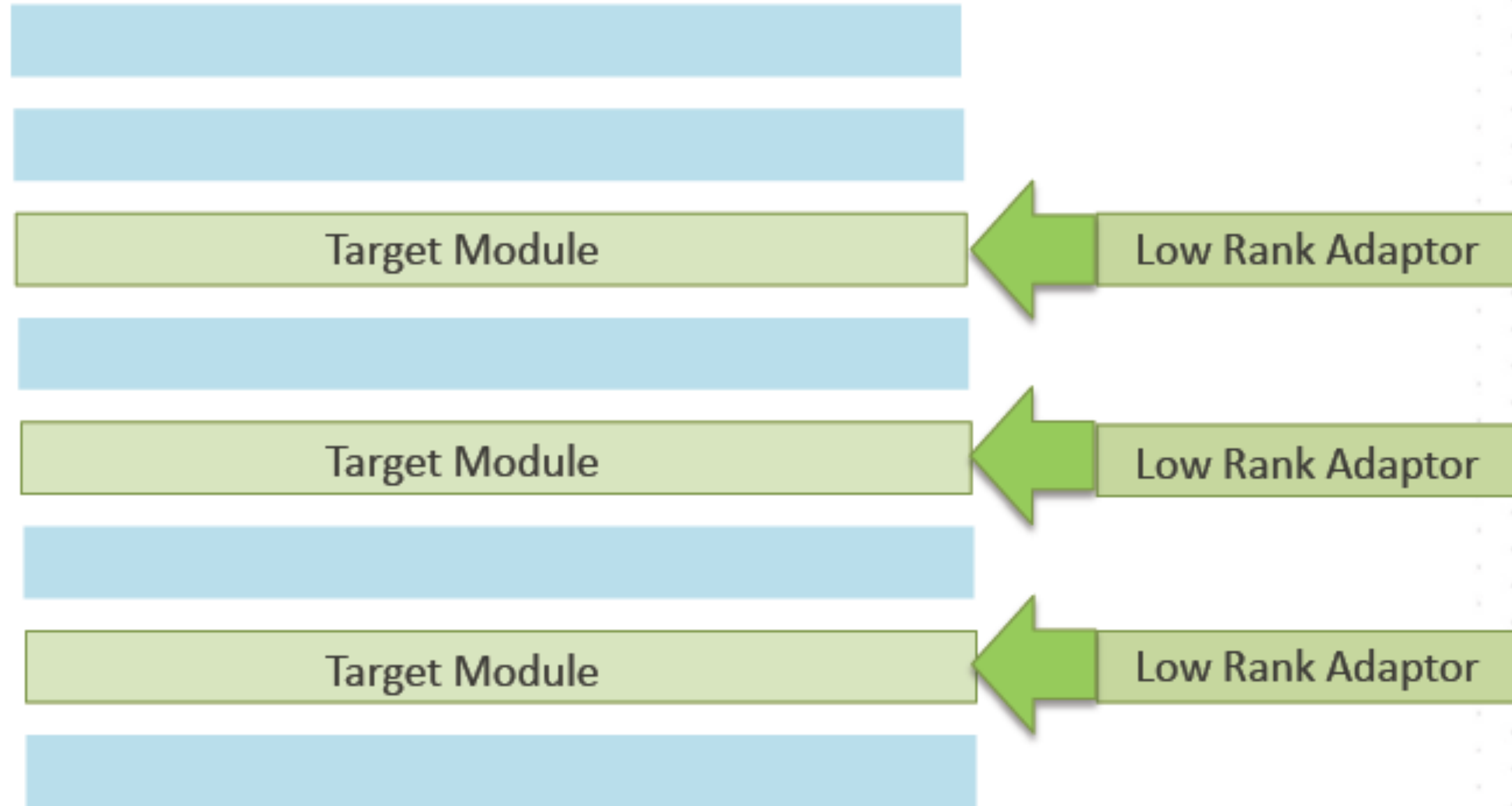


Step 4: Train these and apply them to the target modules



Step 5: Final trick: “Quantization”

Reduce the precision of the Base Model
down to 8 bits or even 4 bits

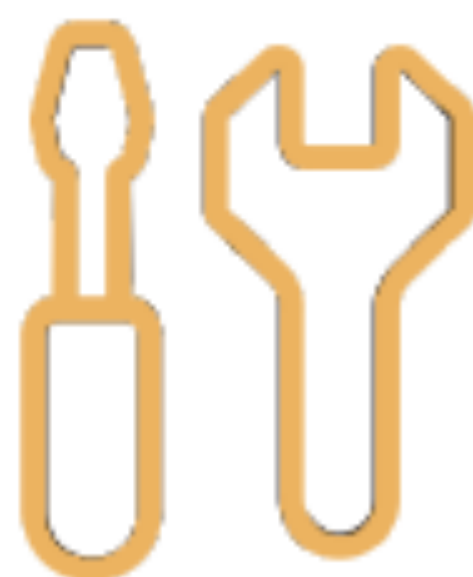


QLoRA = Quantized Low Rank Adaptation

The Hallmarks of an Agentic AI solution



Breaking a larger problem into smaller steps



Using Tools / Function Calling and Structured Outputs



An Environment where Agents collaborate

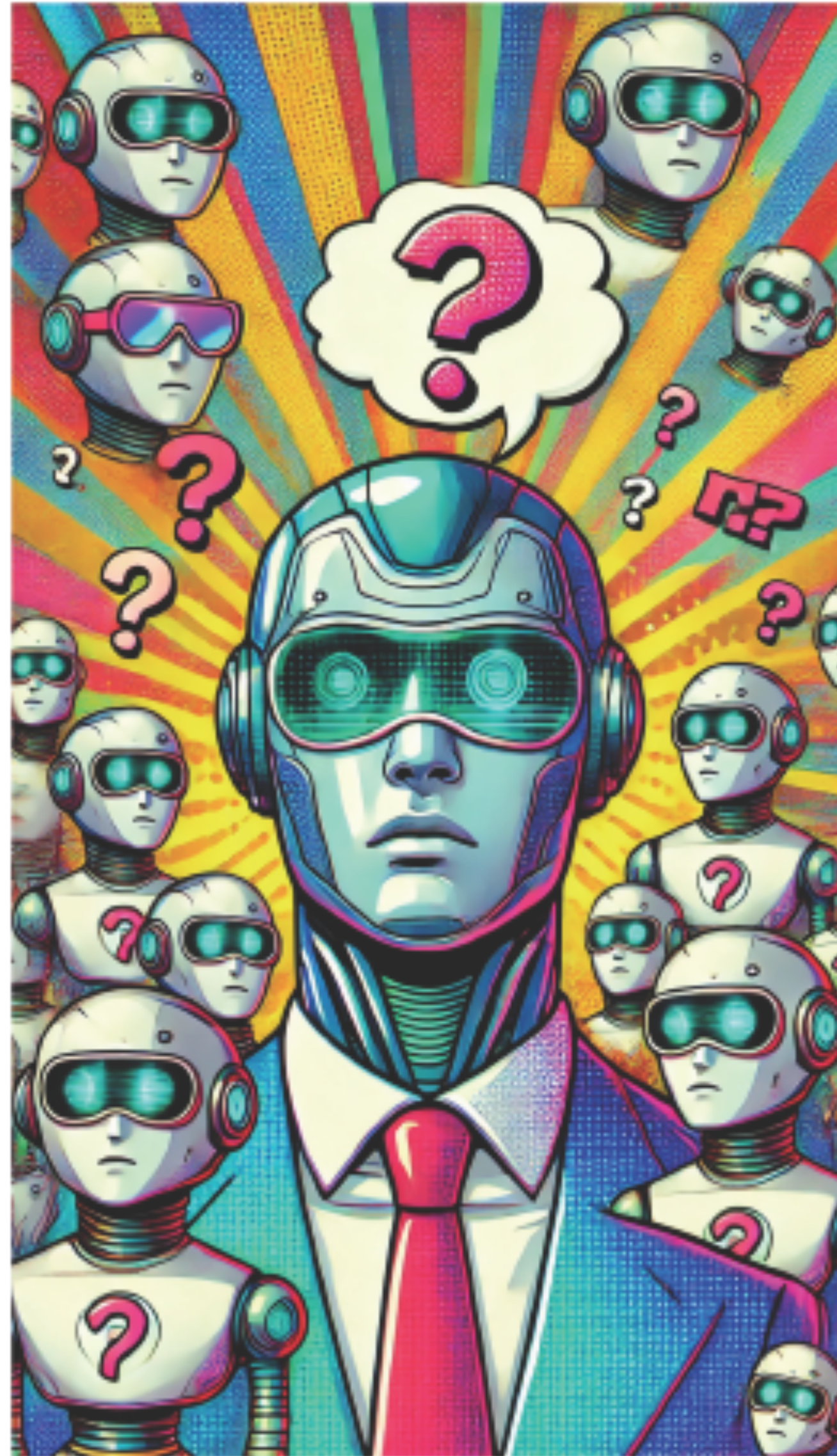


A Planning Agent that coordinates activities



Autonomy & Memory - persists beyond a chat with a human

Decision Points with Agentic AI



- Use an abstraction layer?
 - Lightweight: OpenAI Agents SDK, Crew AI
 - Heavyweight: LangGraph, AutoGen
 - Or, go direct
 - MCP
- How much autonomy
 - Workflow vs Agentic
 - Guard-rails for production

GRAND FINALE

Recap & Beyond The Frontier



Wrapping up

- **Segment 1** reviewed the frontier of LLMs
- **Segment 2** established how to compare LLMs
- **Segment 3** built multi-modal AI assistants, RAG and fine-tuning

Please go back and run the Notebooks in your own time!

*More information linked in the GitHub README.md
to help you directly apply this to your business*

Final extra example projects for those who stayed to the end!



See LLMs try to outwit each other:

<https://edwarddonner.com/outsmart/>

Fine-tuning an LLM on your own Text Message history to create a simulation of yourself!

<https://edwarddonner.com/2024/01/02/fine-tuning-an-llm-on-240k-text-messages/>

Next steps

1. Clone the GitHub repo and run & tweak the examples
2. Review the resources
 - Experiment with the LLMs yourself
 - Visit the Leaderboards and Arena
3. Attend related Live Events

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1. Clone the GitHub repo and run & tweak the examples
2. Review the resources
 - Experiment with the LLMs yourself
 - Visit the Leaderboards and Arena
3. Attend related Live Events

And - identify opportunities to apply LLMs to your business!

THANK YOU!!

Stay in touch:

<https://www.linkedin.com/in/eddonner/>
ed@edwarddonner.com

Please reach out if I can help. Or for advice on selecting the right LLM for your business problem...

You'll be receiving a feedback survey – all feedback is massively appreciated and is instrumental in helping me improve.

